

Belonging, violence, and natural resource governance: experimental evidence from Mali and Niger

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Abstract

The presence of non-state armed groups in the Sahel complicates effective natural resource governance, which is critical for conflict resolution, economic growth, and climate resilience. In response, African governments and donor organizations have alternatively implemented both community-led and state-led interventions to strengthen natural resource governance institutions. It is unclear, however, what governance strategies will be successful in the face of these challenges. We use a vignette survey experiment administered to 3,607 rural household heads in Mali and Niger to show that (1) respondents consistently believe that a community-led natural resource governance system would be more effective, more equitable, and more resilient to climate change than either the status quo or a state-led system; (2) that members of local ethnic outgroups have greater confidence in community-led and state-led regimes relative to the status quo; and (3) exposure to violence reduces confidence in all proposed systems. This paper advances our understanding of natural resource governance under institutional pluralism, explores the endogenous relationship between state capacity and armed conflict, and identifies potential limits on the extent to which strengthening local institutions can substitute for low state capacity. **Keywords:** natural resource governance, political economy, land tenure, survey experiments, Mali, Niger

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Climate variability, environmental degradation, and reduced availability of natural resources pose obstacles to natural resource governance in countries across the Sahel such as Mali and Niger. These challenges compound fragile security situations, heighten competition over water and arable land, and strain pre-existing conflict-resolution mechanisms. In response, international donors and African governments commonly incorporate community-led structures into their natural resource governance interventions (Deininger and Goyal 2024). Such initiatives are predicated on the idea that local and inclusive governance fosters higher levels of trust, legitimacy, and cooperation (Ostrom 2015). However, both state-led and community-led initiatives struggle to cope with these unprecedented pressures. What kind of institutions could improve natural resource management, resolve conflicts, and boost climate resilience in the face of resource strain and ongoing inter-communal violence?¹

This paper characterizes how exposure to conflict, as well as local patterns of belonging and exclusion, condition rural households' expectations of and preferences towards natural resource governance through a vignette survey experiment administered in-person to 3,607 respondents in rural Mali and Niger.² We presented respondents with one of three descriptions of a hypothetical natural resource governance institution. One treatment condition described a "community-based" natural resource system in which communities would make decisions about resource use. A second treatment condition described a "state-led" or centralized natural resource governance regime, in which state officials make decisions about resource use. A third vignette, the control condition, asked respondents to consider the status quo land governance regime. After the vignette was introduced, participants responded to a set of five questions that measure expectations of the assigned system's capacity to manage conflict, fairly allocate natural resources, and increase climate resilience.

Respondents consistently believe that a community-led natural resource governance institution would be more effective than either a state-led natural resource governance system or the status quo. Respondents also preferred a state-led system to the status quo,

1. An anonymized pre-analysis plan is available at the [Open Science Foundation's repository](#).

2. We surveyed 2,081 respondents across the Koulikoro, Koutiala, San, and Ségou regions of Mali and 1,526 respondents across the Maradi and Tahoua regions of Niger.

but less strongly so. Using a respondent’s physical distance from the village centroid as a proxy to measure out-group status, we show that the marginal effect of being treated with either vignette is increasing in distance from the village centroid. These results are surprising and contrary to our pre-registered hypothesis, given that local and community-led institutions often discriminate against members of the local ethnic out-group (called *allochthones* in this context).³ Finally, we hypothesized that exposure to violence would decrease respondents’ confidence in a hypothetical state-led natural resource governance due to the erosion in the state’s monopoly on violence. In line with these expectations, the count of violent events within a 25 kilometer radius of the village erodes confidence in both state-led and community led natural resource governance. These results suggest that both belonging—alongside corresponding systems of exclusion—and violence condition preferences for natural resource governance.

Natural resource governance across Africa often exists in a state of institutional pluralism: households often have multiple forums to which they could present a dispute (Lust 2022). Existing research suggests that households in the developing world perceive informal institutions to be more fair or less burdensome (Acemoglu et al. 2020; Chaara, Falisse, and Moriceau 2022; Winters and Conroy-Krutz 2021). Normative regimes often overlap when it comes to land tenure. New institutions may add layers to overlapping regimes rather than simplify them (Lund 2008). This pluralism gives households the ability to ignore institutions that they perceive to be illegitimate or ineffective. Therefore, identifying the factors that drive households’ confidence in natural resource governance institutions is critical to understanding how such institutions may or may not contribute to economic growth or peacebuilding.

Paris (2004) argues that providing effective institutions is a necessary first-step for broader peacebuilding. Likewise, a broad literature clarifies the endogenous relationship between state capacity and development (Besley and Persson 2010; Blattman and Miguel

3. *Autochthones*—members of the local in-group—exert a customary claim to the land, often derived via shared descent from the community’s founding lineage (Colin, Kouamé, and Soro 2007; Boone et al. 2021; Vlavonou 2023). *Allochthones* are later arrivals and are often excluded from village institutions. Elsewhere, *allochthones* are referred to as ‘strangers,’ ‘newcomers,’ ‘local outsiders,’ or ‘tutees’ (in contrast to their ‘tutors,’ the *autochthones*). We prefer “autochthone” and “allochthone” as terminology to capture a relationship which is distinctive in its particulars yet common to much of West Africa.

2010). An absence of conflict resolution institutions, and the consequent lack of stability, hampers any broader political reform which could address the root causes of conflict. One solution to this problem has been to use community-based institutions to ‘bootstrap’ peace (Fearon, Humphreys, and Weinstein 2009; Hartman, Blair, and Blattman 2021). These approaches are predicated on local institutions maintaining higher levels of trust and legitimacy, relative to top-down governance (Ostrom 2015). This paper uncovers potential challenges to this approach: violence which undermines state institutions may also render a pivot to community-based interventions ineffective. Where ongoing violence degrades trust in both the state and in community institutions, rural citizens may find themselves in an authority vacuum. By illustrating this potential spiral, our findings identify potential challenges to correctly sequencing peacebuilding efforts.

More broadly, these results have consequences for literature which suggests that community institutions can substitute when state institutions or capacity fails. Institutions are ‘the rules of the game’ on which citizens mutually agree (North 1990; Callen, Weigel, and Yuchtman 2024). Recent literature has shown that strengthening local institutions’ ability to resolve disputes led to reductions in violence (Hartman, Blair, and Blattman 2021; Christensen et al. 2025; Ribar et al. 2025). However, when citizens lack confidence in an institution, they will not rely on it. We show that exposure to violence—across all specifications and outcome variables—decreases respondent’s confidence in both community and state-led natural resource governance. This finding places a scope condition on a growing literature which posits that that community institutions can substitute when state capacity fails (Van Der Windt et al. 2019; Balán et al. 2022; Greico 2025) by illustrating the conditions under which rural households believe that ‘improved’ institutions would be effective.

Informal natural resource governance in West Africa often excludes relative newcomers such as allochthones and pastoralist groups from village governance (Delville and Moalic 2019; Boone et al. 2021). The descendants of the first settlers of the village, called autochthones, generally dominate village governance. The village politics surrounding belonging and exclusion creates heterogeneous returns to participating in village land governance. Low-status members of the in-group or members of the out-group are likely to be dissatisfied with land governance (Funjika and Honig 2024). Our

paper advances this literature on the local politics of belonging and exclusion with a surprising finding: members of the local outgroup are so dissatisfied with the status quo that they evaluate idealized institutional designs on their own terms rather than against historical experience. More broadly, the paper shows how two factors—belonging and violence—condition how much respondents think natural resource governance institutions could resolve conflicts, secure property rights, and improve conflict resilience.

I Context: land governance in the Sahel

Natural resource governance in the Sahel chiefly concerns access to land and access to water. Land in Africa—including in Mali and Niger—is often but not always commonly held. Hughes (2014, 10) notes “three different types of lands [within villages]: individual lands, family lands, and village common lands also known as chieftaincy lands.” Land near villages is used to grow crops such as cotton, rice, or sorghum; smaller plots are used for market gardening vegetable crops such as tomatoes or onions. Agriculture is generally rain-fed and seasonal. Both Mali and Niger also have large transhumant populations; these pastoralists travel with their herds in search of pasture and water. However, increased density of agriculture has encroached traditional livestock corridors which connect pastures and provide livestock with access to water (McGuirk and Nunn 2025).

Land tenure security and effective natural resource governance is critical to both economic growth and sustainable peacebuilding in rural Africa. Land tenure security conditions households’ ability to make investments: households are unlikely to fertilize their parcels, build irrigation, or plant trees if they are unclear if they will reap the benefits (Goldstein and Udry 2008). Secure land tenure helps minimize the outbreak of natural resource disputes by clarifying boundaries and providing easier access to conflict resolution institutions. Effective natural resource governance is also crucial for climate resilience: secure land tenure can help reduce deforestation and encourage conservation (Deininger and Goyal 2024, 183–217).

The status quo of natural resource governance in Mali and Niger—what we reference in the control condition of our experiment—reflects both low state capacity and the endurance of customary institutions (Winters and Conroy-Krutz 2021). Most natural

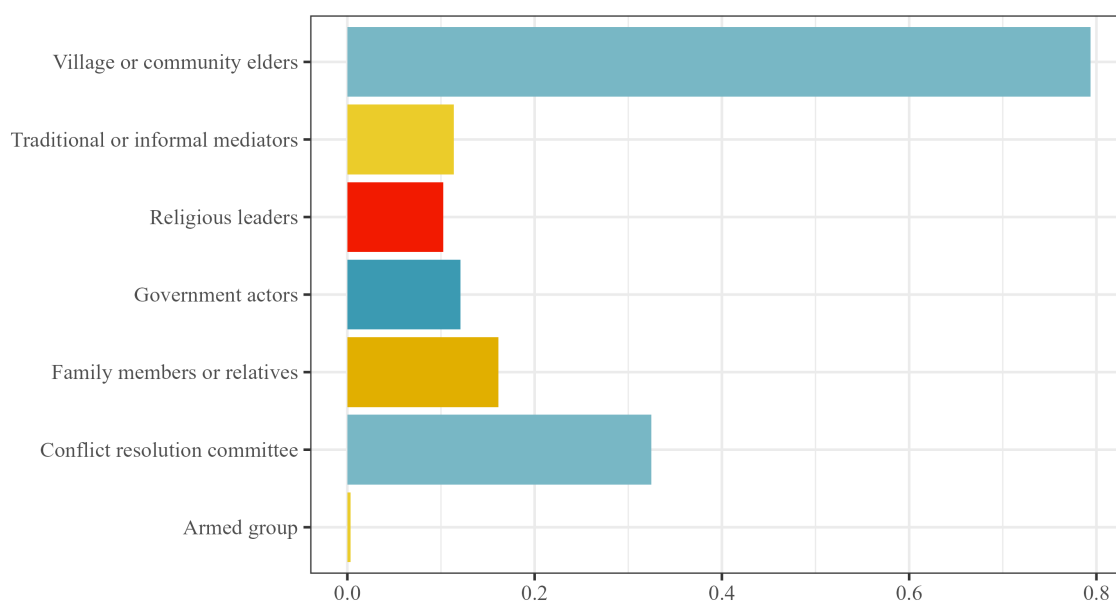
resource governance takes place within customary institutions, rather than formal institutions of the state. In Niger, the relevant state institutions are land commissions (*Commissions Foncières*) which operate at the department level (second-level administrative division), but are understaffed and under resourced. In addition, land commissions are not empowered to resolve land disputes; they can only issue titles for undisputed land (Hughes 2014, 13). Governance is similar in Mali: the country launched a new framework for land governance in 2000, the *Loi Domaniale et Foncière*, but in practice the state remains scarce. According to the 2021 wave of the Living Standards Measurement Survey (LSMS), only three percent of households in Mali and five percent of households in Niger possessed a formal land title for at least one of their agricultural parcels.

The combination of low state capacity and strong customary institutions creates a scenario in which institutional pluralism predominates: rural households have multiple forums to which they could refer a dispute. Before administering our experiment, we asked respondents to our survey “If you have a dispute about land, livestock, or a business transaction, who would you approach to resolve the dispute?” Figure 1 shows these results. The overwhelming majority of respondents (2,861) said that they would approach a village or community elders. However, of the 3,607 respondents, 45 percent selected two or more responses and 13.5 percent selected three or more answers. In other words, land governance in Mali and Niger is an institutional context where there are multiple sets of ‘rules’ which respondents may reference when attempting to resolve a dispute (Callen, Weigel, and Yuchtman 2024). In such a context, shifting opinions towards which institution is more legitimate or effective can meaningfully change the institutions on which citizens rely to resolve their conflicts.

Absent formal land governance, customary institutions (most often in the form of village chiefs or community elders) adjudicate disputes. In both Mali and Niger, customary access to land is predicated on the ‘right of the axe,’ or land rights awarded through descent from the original settlers of the village (Hughes 2014).⁴ These customary land rights distinguish the autochthones, or descendants of the original settlers of the village, from allochthones, or more recent arrivals. More recent arrivals to the village are generally

4. The village chief almost universally descends from this lineage; the name comes from the act of clearing the bush.

Figure 1. Preferred institutions for dispute resolution in Mali and Niger



This figure shows responses to “If you have a dispute about land, livestock, or a business transaction, who would you approach to resolve the dispute” from the authors’ survey in Mali and Niger. Respondents were able to select multiple answers.

granted long-term use rights to land, rather than land ownership. Many allochthones stay in villages for generations while paying debatably ceremonial rents for land. Tensions often emerge as to the extent that these rents are actually ceremonial—particularly when the following generations of allochthones ‘inherit’ the parcels. Ethnic groups that are predominantly herders rather than farmers—such as Tuaregs and Peulhs (Fulani)—are almost always considered allochthones. Other ethnicities would also be considered allochthones outside of their village of origin. Most Tuaregs and Fulani are allochthones, but that does not imply that most allochthones are Tuaregs and Fulani.⁵

The paucity of natural resource governance allows conflict to fester. Many conflicts in sub-Saharan Africa have their genesis in competition over resources. When we asked respondents to name to two most common causes of clashes or violent conflicts, 67 per-

5. This paragraph describes autochthony and allochthony in Mali and Niger. These groups could be considered autochthones in other locations. Peulhs, for example, are the ethnic majority in Guinea.

cent cited natural resources such as land or water. In Mali, Benjaminsen and Ba (2024) narrate how a conflict over access to riparian pastures in Mali led to Katiba Macina, a local offshoot of Al Qaeda, intervening on behalf of herders. Likewise, land tenure conflicts contributed to the outbreak of recent conflicts in both Côte d’Ivoire and the Democratic Republic of the Congo (Boone 2014; Autesserre 2010). Such conflicts are likely to increase in both severity and frequency in the Sahel as climate change and desertification make access to water scarce and shrink the pool of arable land. Ongoing violence persists in both Mali and Niger, as the state has been unable to conclusively defeat insurgent actors. The porosity of borders in the Sahel means that the same violent groups—chiefly Jama’at Nusrat al-Islam wal-Muslimin (JNIM) and the Islamic State in the Greater Sahara (ISGS)—operate in a broad theater encompassing both countries, as well as Burkina Faso.

2 Belonging, violence, and natural resource governance

Given these challenges, policymakers and international donors have proposed a variety of potential interventions to improve natural resource governance in rural Africa. Many African governments have taken land tenure into their own hands and implemented state-led natural resource governance. In Rwanda, a top-down effort led to approximately 11.5 million parcels of land being registered between 2011 and 2013 by the National Land authority (Deininger and Goyal 2024, 60). Ali, Deininger, and Duponchel (2017) show that this process had positive effects on both gender equality and access to credit. Elsewhere, the Liberia Land Authority provides a more recent example of such a state-led natural resource governance program; it advertises itself as a ‘one-stop shop’ for land titling. These efforts aim to shift the institutional equilibrium such that citizens rely on the state, rather than on local community institutions, for land governance.

Other initiatives have emphasized the need for locally-led natural resource governance programs. Such programs date back to the late 1980s, when the World Bank and other donors piloted the Plans Fonciers Rurale (PFR) programs in Côte d’Ivoire, Benin, and Burkina Faso (Delville and Moalic 2019). More recently, Hartman, Blair, and Blattman (2021) show that introducing alternative dispute resolution in Liberia led

to a decrease in violence driven by land disputes. In Niger, layering conflict resolution training on top of a livelihoods program for youth decreased both support for violence in treated villages and the count of violent incidents (Ribar et al. 2025). These programs suggest that community-led natural resource governance programs may substitute to protect property rights and mitigate conflicts even when state capacity is low (Balán et al. 2022; Greico 2025; Van Der Windt et al. 2019).

Natural resource governance is under-provided in both Mali and Niger. Nevertheless, effective natural resource governance is key for economic growth, conflict resolution, and climate resilience. Given the problems associated with the status quo, it seems likely that households will perceive any alternative to the status quo to be a more effective form of natural resource governance. With this background in mind, we hypothesize that:

H.1a Respondents will prefer either community-based natural resource governance or state-led natural resource governance to the status quo.

Respondents' discontent with the status quo does not imply ambivalence between the alternatives. Winters and Conroy-Krutz (2021, 2) argue that “[c]itizens might assess that informal institutions, which are typically rooted in traditional authority and customs, have significant legitimacy and are well-tailored to local contexts. Formal institutions, on the other hand, rely on rules and procedures often developed in distant power centers and rooted in Western legal traditions, and might therefore seem foreign or even illegitimate.” Using a survey experiment in Mali, they show that respondents think that a formal institution would produce a less fair result (albeit one more in accordance with the formal laws of Mali), and more likely to require a payment.

These dynamics are not unique to the Sahel. In Burundi, a legal aid program increased the uptake of state justice institutions, particularly among members of marginalized communities (Chaara, Falisse, and Moriceau 2022). However, the program did not make participants less likely to use local, informal forums and did not increase participants' trust in formal justice. Funjika and Honig (2024) show that individual status within both formal and informal justice institutions conditions the perceptions of conflict resolution institutions. Nevertheless, respondents are most likely to take their disputes to a customary forum even when a state-led or statutory forum is available. The argument that

community based governance engenders higher levels of trust, legitimacy, and cooperation (Ostrom 2015) motivates a growing literature on using community-based institutions to strengthen peace (Fearon, Humphreys, and Weinstein 2009; Hartman, Blair, and Blattman 2021). Consequently, we further hypothesize that:

H.1b Respondents will prefer community-based natural resource governance to state-led natural resource governance.

Villages in Mali and Niger are rarely homogeneous; rather, they are split between “natives” and “strangers.” Natives, more formally called autochthones, descend from the initial settlers in the area and usually hold some customary claim over a village’s land. Outsiders, or allochthones, are descendants of later settlers, and are often but not always a different ethnicity than the autochthones. Colin, Kouamé, and Soro (2007, 34) note that this system “perpetuates a patronage relationship between autochthons [sic] and strangers (in the sense of ‘non locals’), to whom rights in land are extended... The migrant owes his tuteur a perennial gratitude (transferred to his heirs), expressed through gifts of agricultural products, contributions to his tuteur’s expenses at times of funerals, and so forth.”

The ethnic heterogeneity which characterizes many villages in West Africa further complicates natural resource governance. Existing institutions often do not benefit populations equally, with members of the customary in-group benefiting more than the customary out-group (Funjika and Honig 2024). Differential returns to participating in local institutions can shift both the confidence in these institutions and the eventual up-take of these institutions. For example, Acemoglu et al. (2020) show heterogeneous up-take of state courts in Pakistan by caste.

Claims of autochthony serve to control access to land or resources; Boone (2014, 329) illustrates that “[s]ocial hierarchy, cleavage, and exclusion within these local arenas are often defined by control over agricultural land and labor, and access to water and pressure.” When land became more scarce, autochthones began to rely more heavily on their customary claims to land. In Ghana, Berry (2009, 40) notes that “intersecting tensions over eligibility for land access... contributed to a resurgence of appeals to ‘tradition’ and historical precedent to validate claims to land and citizenship.” As land becomes more

valuable, ethnicity can become increasingly salient as a rural landowners use it to leverage access to resources (Pengl, Roessler, and Rueda 2022). The relationship between autochthone and allochthone may change over time. Allochthones have begun to outnumber autochthones in certain areas of Côte d’Ivoire, due to large influxes of migrants encouraged during the presidency of Félix Houphouët-Boigny (Boone et al. 2021). These demographic changes can happen both at the regional level, but also at the village level, as allochthonous settlements occasionally eclipse their “hosts village” in size or economic prominence. Consequently, when chiefs in Côte d’Ivoire are able to capture village-level natural resource governance, they generally exclude allochthones from access to formal property rights (Ribar 2025).

Autochthones tend to dominate village politics, leaving allochthones in a “subaltern social position” (Delville and Moalic 2019, 332). In Mali and Niger, chiefs are generally descendants of the initial settlers of the land (Hughes 2014). Weak state capacity in many peripheral areas cements the role of these elites: where early state interventions created inequality which benefited the autochthone elite, the inequalities are more likely to persist where the state does not subsequently intervene or provide services (Nathan 2023). Even today, rural allochthone settlements depend administratively on villages populated by autochthones. These dynamics mean that allochthones may perceive these local institutions to be biased against them. In other words, an allochthone in rural Mali or Niger may prefer the ineffective governance of a scarce state to a local governance that is biased against them.⁶

Community-led natural resource governance opens space for the capture of land tenure institutions by existing elites and decisions which are biased against households with low-status in these institutions, such as ethnic out-groups. It follows that allochthones will expect a community-led natural resource regime to reflect existing constellations of power within the village and continue the status quo of discrimination in natural resource governance. A state-led natural resource governance regime, however, would be less susceptible to these local dynamics. For these reasons, we hypothesize that:

6. On the other hand, African states are not passive bystanders when it comes to autochthony—rather, they have often promulgated policies which support autochthones access to land to shore up electoral support (Berry 2009; Boone et al. 2021).

H.2a Allochthonous respondents will have higher support for state-led natural resource governance, relative to autochthonous respondents.

H.2b Autochthonous respondents will have higher support for community natural resource governance relative to allochthonous respondents.

Allochthones may be inherently skeptical of any community-led intervention which purports to be inclusive. Consequently, H.2A suggests that allochthones would not believe the portion of our community-led vignette that suggests “meetings are inclusive, and decisions are made by consensus.” It is also possible that allochthones take the vignette at face value and believe that the proposed natural resource governance regime would be inclusive. Whether respondents believe in the inclusivity of the proposed regime is ultimately an empirical question; we provide suggestive evidence that the effect of receiving both vignettes is weaker among allochthones exposed to a ‘proof of concept’ for inclusive natural resource governance via the Justice and Stability in the Sahel (JASS) program we describe below, relative to allochthones not exposed to JASS. Though these results are not pre-registered, they suggest that allochthones may simply be disillusioned with their status quo, making them willing to support any alternative.

Beyond questions of belonging and exclusion, we also expect that exposure to violence will reduce the perceived effectiveness of state-led natural resource governance. The role of violence in decreasing the perception of state capacity is well established in other contexts. In Afghanistan, households exposed to greater levels of violence view the police (who may enforce property rights) as less effective and less procedurally fair (Deglow and Sundberg 2021). Gates and Justesen (2020) show that an unexpected rebel attack in Mali halfway through a public opinion survey decreased trust in the president, although this effect did not extend to all government offices.⁷

The negative relationship between exposure to violence and trust in institutions is likely to be even stronger when it comes to property rights. Natural resource conflicts

7. In contrast, Harding and Nwokolo (2024) show an increase in general political trust following unexpected attacks by Boko Haram in Nigeria. Unlike Harding and Nwokolo, however, we do not study a one-off set of attacks but rather a long-term pattern of violence. Our study also explores perceptions of state capacity, rather than political trust per se.

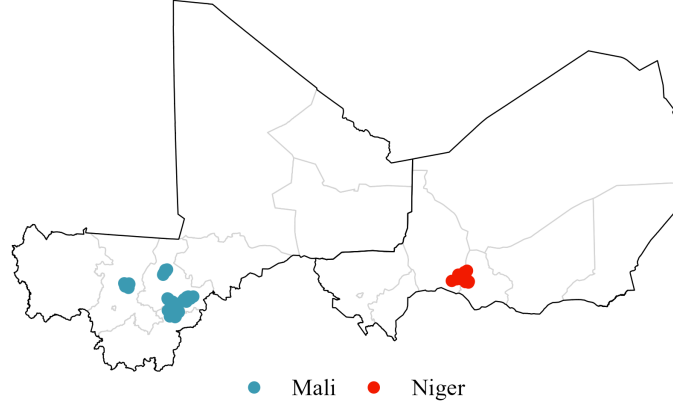
and land disputes are a key enabling factor for the spread of violence across the Sahel (Raleigh, Nsaibia, and Dowd 2021). Across Africa, decreased rainfall within areas traditionally used by pastoralist communities leads to increased pressure on land and water in neighboring areas inhabited by farming communities, leading to increased conflict (McGuirk and Nunn 2025). Armed groups in the Sahel often leverage local natural resource conflicts to justify their interventions. A jihadist organization in Mali, for example, escalated a conflict over pastoralist access to riparian pastures (Benjaminsen and Ba 2024); where state-led natural resource governance institutions began to extract rents from pastoralists, the jihadist groups intervened on behalf of a population it perceived to be its constituents.

When respondents stop seeing the state as an effective provider of security, rural citizens may no longer see it as a legitimate source of conflict resolution or property rights. Enforcing property rights necessarily requires the capacity to exclude non-rights holders from the property. Without a monopoly on violence, landholders may reasonably doubt that the state has such a capacity. Viewing the state's inability to reduce violence does not require extrapolating an inability to enforce property rights; rather, ongoing violence directly reveals the state's incapacity to provide effective natural resource governance. We hypothesize that:

H.3 Respondents with greater exposure to violence will have lower support for state-led natural resource governance, relative to respondents with less exposure to violence.

Because institutional pluralism means citizens have a choice to bring disputes or land conflicts to multiple institutions, citizens need not bring disputes to an institution which they do not trust. The state's inability to resolve natural resource disputes produces the potential for an institutional spiral: conflict decreases the state's ability to provide dispute resolution in the hinterlands, the resulting disputes metastasize into violence, and the violence further undermines the confidence in the state. Factors such as climate stress or resource scarcity may compound this negative equilibrium. This endogenous relationship between conflict and state capacity means that conflict may prevent natural resource governance interventions from meaningfully shifting citizens' confidence in their institutions, and thereby the institutions on which citizens rely.

Figure 2. Locations of surveyed villages in Mali and Niger



This figure shows the 24 villages in Niger and 56 villages in Mali in which we implemented the field survey.

3 Research design and methodology

3.1 Sampling and survey implementation

To explore these hypotheses, we administered a survey experiment to 2,081 rural household heads in Mali and 1,526 in Niger. We collected data in the Koulikoro, Koutiala, San, and Ségou regions of Mali, as well as the Tahoua and Maradi regions of Niger. Figure 2 shows the villages in which sampling took place.

Our survey takes place within a broader Foreign and Commonwealth Development Office (FCDO)-funded program, titled “Justice and Stability in the Sahel” (JASS).⁸ JASS was implemented through two stages. The first stage (2021 to 2023) emphasized ‘fragile yet stable’ buffer zones in south-central Mali. JASS carried out scoping missions and consultations with administrative, civil society, religious, and traditional leaders, as well as national authorities; this participatory process selected 12 communes for im-

8. The Washington University in St. Louis IRB approved the secondary usage of these data for research under protocol number 202511107.

plementation. During phase two of JASS implementation, JASS expanded to Niger’s Tarka Basin (communes in Maradi and Tahoua) as well as eight new communes in Mali. Within each commune, local selection committees composed of representatives of all key social groups, chose specific target villages and participants. The committees applied jointly developed criteria that emphasized: representation of different population groups (ethnic, gender, youth, traditional authorities); vulnerability to resource-based conflicts and justice gaps; accessibility and security feasibility for program delivery; presence of governance institutions that still had some legitimacy; strategic importance (population density, resource pressures, transhumance routes). We did not survey within communes where less than ten villages received the JASS treatment.

We use village fixed effects in all regressions to absorb the direct effects of the JASS intervention. We calculated sample sizes to ensure adequate coverage of a series of monitoring and evaluation indicators for the broader JASS program, rather than for the survey experiment specifically. However, our pre-analysis plan shows minimum detectable effects of 0.237 and 0.315 standard deviations of the outcome variables, below the magnitude of the effects we find in tables 2 and 3.

Within sampled villages, enumerators conducted a random walk to select respondents to which to administer the survey. Enumerators administered an informed consent statement before beginning the survey. Because of the idiosyncratic sampling strategy, we do not weight our regressions. We consider these results to indicative of areas where state institutions are largely absent and land conflicts remain common, rather than the broader population of Mali or Niger. This population is the set of respondents for whom natural resource governance remains most challenging and at whom similar interventions have been targeted.

3.2 Survey measures

We use a vignette survey design, in which we presented respondents with one of three descriptions of natural resource governance institutions. Respondents randomly received the community treatment with probability 0.25, the centralized treatment with probability 0.25, and the control condition with probability 0.5. The three treatments were:

1. **Community treatment:** As you know, communities are facing challenges in managing natural resources (e.g., water, forests, land) due to the impacts of climate change, such as droughts, floods, and soil degradation. Imagine that to address these challenges, your community has established a **community-based natural resource governance system**. Under this system, local stakeholders, including farmers, community leaders, landowners, and resource users, meet regularly to make decisions about resource management. These meetings are inclusive, and decisions are made by consensus. The system emphasizes cooperation, and conflicts are addressed through local mediation and dialogue. External organizations, such as NGOs, provide support by offering training on sustainable practices and climate adaptation strategies, but the community makes the decisions about resource use.
2. **Centralized treatment:** As you know, communities are facing challenges in managing natural resources (e.g., water, forests, land) due to the impacts of climate change, such as droughts, floods, and soil degradation. Imagine that to address these challenges, your community has established a **centralized, government-controlled natural resource governance system**. Under this system, decisions about resource management (e.g., water allocation, forest use, land management) are made by government agencies at the national or regional level. The government sets regulations and policies for how resources are used and enforces them with support from law enforcement. Local communities have little input in the decision-making process, though they may receive financial incentives or technical support from the government to help adapt to climate change. Conflicts over natural resources are resolved through legal channels or government-appointed arbitrators, rather than community dialogue.
3. **Control:** As you know, communities are facing challenges in managing natural resources (e.g., water, forests, land) due to the impacts of climate change, such as droughts, floods, and soil degradation. Imagine that your community continues to manage natural resources in the same way it has been doing for years.

After enumerators read one of these three treatments aloud, respondents answered

five questions. These outcome variables include:⁹

1. Do you think there would be an increase in **resilience** to climate-related challenges (e.g., droughts, floods, land degradation) within the community? [yes/no]
2. Do you think the system will **reduce conflicts** over natural resources (e.g., water, land, forests) in the community? [yes/no]
3. To what extent do you trust that land-related issues will be managed **fairly** and transparently? [Not at all/a little/neutral/somewhat/completely]
4. Do you believe that the resolution of conflicts/disputes will be **equitable**? [yes/no]
5. Do you believe that the authorities will voice concerns and make decisions that **benefit all members** of the community? [yes/no]¹⁰

We designed the “status quo” (control) vignette in collaboration with Mercy Corps’ Mali and Niger country offices as a minimal baseline representing respondents’ existing governance arrangements. This existing arrangement comprises a mixed local-led and state-led system, in which a theoretically top-down property rights regime co-exists with the presence of customary elites that manage day-to-day land tenure (Hughes 2014). As we describe above, the local portion of this regime is often non-inclusive, excluding marginalized groups or advancing the interests of local elites (Lund 2008; Boone 2014).¹¹ Consequently, natural resource governance in Mali and Niger (and much of Africa) exists somewhere on a spectrum between state-led and community-led natural resource governance.

We designed the two treatment vignettes as the logical end-points to this spectrum of land tenure. They contrast (1) an inclusive community-led governance model emphasizing regular participation by multiple stakeholder groups and consensus-based decision-

9. We use the words in bold as shorthand for these outcomes in tables 2 and 3.

10. These outcome variables are all strongly and positively correlated with each other. Between these outcomes, the coefficients of correlation range from 0.56 to 0.86.

11. This co-occurrence is not accidental; local elites often resist the expansion of formal property rights because customary regimes provide local elites with a reservoir of political authority, even while gatekeeping access to formal property rights provides opportunities for rent-seeking (Ribar 2025).

making with (2) a centralized state-led model emphasizing formal regulation and enforcement by government authorities. The community-led and state-led treatment vignettes do not precisely mirror each other. Rather, the vignettes are meant to reflect the universe of real world programs, which often bundle property rights interventions with strengthening community deliberation mechanisms (Hartman, Blair, and Blattman 2021; Christensen et al. 2025) or with state-led interventions on areas like climate resilience or forestry (Huntington and Shenoy 2021).¹²

To test hypotheses 2.a and 2.b, we also need to identify whether a respondent is an autochthone or an allochthone. We pre-registered two measures to capture autochthony. First, we ask respondents their status within the village: whether they were a "host," or if they had moved to the village, returned to the village, or were a refugee. We code hosts as autochthones and all other respondents as allochthones.¹³ We also measure allochthony through ethnicity—specifically, by coding Peulh/Fulani and Tuareg respondents as allochthones. This coding reflects these ethnicities' status in Mali and Niger as herders who often transit between villages and are treated as outsiders. However, both strategies to capture the allochthone/autochthone divide leave us underpowered to detect differences. When asking respondents directly if they are a host or a more recent arrival, only 69 respondents report being an allochthone. Similarly, only 282 respondents are either Peulh/Fulani or Tuareg.¹⁴

To overcome these challenges, we code a third measure of allochthony: distance between the village's centroid and the respondent's dwelling.¹⁵ This measure captures how far a respondent lives from the center point of the village. Initial arrivals to a

12. For an overview of the evidence surrounding such interventions, see: Huntington and Lisher 2025.

13. Hosts or tutors is a common way to refer to the original settlers of the land, who in the conventional narrative then permit later arrivals, or allochthones, to settle.

14. One alternative could be to code whether a respondent belongs to the majority ethnicity within the village, as measured by our survey. However, village demography does not always cleanly map to ethnicity. In-migration can create scenarios where autochthones become minorities in their own villages while maintaining their political authority (Boone et al. 2021). Autochthony is also generally a function of lineage, rather than ethnicity: a respondent could be considered an allochthone while belonging to the same ethnicity as a village's initial lineage. Nevertheless, appendix table A11 shows no heterogeneous effects by whether a respondent belongs to the village's ethnic majority.

15. The respondents' location is captured during the survey via GPS; we calculate the village centroids by averaging these GPS coordinates by village.

village—the autochthones—generally create a central settlement. They farm around this central point, creating a band of farmland surrounding the village. Later arrivals—allochthones—often settle in hamlets outside of these initial bands of farmland, where land remains available (Delville and Moalic 2019). Villages may also settle allochthones strategically on the outskirts of their territories to shore up territorial claims (Delville and Moalic 2019; Ribar 2025, 33). To avoid our results being driven by outliers or GPS errors, we winsorize these results to be within the 5th and 95th percentiles. We then take the inverse hyperbolic sine of the distance measure to normalize it (Bellemare and Wichman 2020).¹⁶ While GPS measurements may contain errors, table A13 in the appendix shows substantively identical results when sub-setting to villages where the village centroid is within 500 meters of a village’s center point identified by hand using satellite imagery.

For hypothesis 3, we need to measure respondents’ exposure to violence. In our survey, we asked respondents “To your knowledge, have there been any violent confrontations or conflicts in your community in the last six months?” However, we expect this question to be vulnerable to either social desirability bias or a general unwillingness to discuss violence. As a result, we use the count of ACLED events within a 25 kilometer radius of the village as a measure of exposure to violence. We exclude ACLED events which are coded as protests and events where the primary actor is government forces, as these do not match our theoretical explanations of an incomplete monopoly on violence. We calculate these statistics for the one year prior to the survey, but appendix A.2 shows these results calculated using two and three year pre-treatment periods, as well as versions calculated using 10 kilometer radii. Following Bellamare and Wichman (2020), we use an inverse hyperbolic sine transformation to normalize these data. We did not specify such a transformation in the pre-analysis plan; however, table A.2 shows that results are substantively identical with and without this transformation and across essentially all specifications.¹⁷

For control variables, we include sex, age, age squared, and a set of binary indicators

16. Appendix tables A9 shows qualitatively similar results to distance using self-reported autochthony; table A10 shows no heterogeneous results by whether a respondent belongs to one of the herding ethnicities.

17. Table A3 in the appendix shows no heterogeneous effects of treatment by self-reported encountering violence. However, exposure to ACLED events is indeed higher among respondents who reported they experienced violence.

for the respondent participating in any specific JASS intervention. Table 1 shows the balance between the difference treatment groups. Respondents who received the three treatment items are balanced on all characteristics except for participating in infrastructure initiatives as part of JASS. With 38 t -statistics in table 1, this imbalance is consistent with a successful randomization, and is not a cause for concern.

Table 1. All three treatment groups are balanced across control variables

	Control		Local-led Treatment			State-led Treatment		
	Mean	Std.Err.	Mean	Std.Err.	T-score	Mean	Std.Err.	T-score
Age	42.71	15.06	42.66	14.19	-0.09	42.76	14.25	0.09
Sex (male)	0.58	0.49	0.61	0.49	1.78	0.58	0.49	0.11
Ethnicity								
Bambara	0.25	0.43	0.26	0.44	0.67	0.26	0.44	0.56
Haoussa	0.40	0.49	0.37	0.48	-1.34	0.39	0.49	-0.44
Mandé	0.04	0.19	0.03	0.17	-1.02	0.03	0.17	-0.76
Minianka/Sénoufo	0.14	0.35	0.16	0.37	1.50	0.15	0.36	0.89
Other	0.04	0.19	0.04	0.19	0.23	0.04	0.20	0.57
Peulh	0.03	0.17	0.04	0.21	1.69	0.04	0.19	0.61
Soninké	0.06	0.25	0.05	0.22	-1.35	0.05	0.23	-1.07
Touareg	0.05	0.21	0.04	0.20	-0.34	0.04	0.19	-0.94
Participation in JASS activities								
Awareness	0.39	0.49	0.37	0.48	-1.38	0.38	0.48	-0.85
Capacity building	0.20	0.40	0.19	0.39	-0.76	0.22	0.41	0.90
Infrastructure	0.12	0.33	0.09	0.29	-2.24	0.12	0.33	-0.03
Peace initiatives	0.20	0.40	0.19	0.39	-1.10	0.22	0.41	0.77
Non-farm livelihoods	0.13	0.34	0.12	0.32	-1.25	0.14	0.35	0.58
Agriculture	0.25	0.43	0.23	0.42	-1.00	0.25	0.44	0.42
Resource management	0.10	0.31	0.10	0.30	-0.14	0.13	0.34	2.00
Climate resilience	0.10	0.30	0.11	0.31	0.31	0.10	0.30	-0.48
Advocacy	0.03	0.18	0.02	0.15	-1.38	0.03	0.18	0.03

Note: This table shows the balance of control variables across the three treatment groups from the JASS endline survey. T-scores are calculated using the control group as a baseline.

3.3 Estimation strategy

We use a series of ordinary least squares regressions to estimate these results. All regressions include village-level fixed effects for two reasons. First, our survey experiment was

embedded in a monitoring and evaluation effort for a stabilization and conflict resolution program; our fixed effects absorb exposure to this program. Second, we are interested in how the experimental treatment shifts preferences for natural resource governance, rather than how a village’s conditions shift preferences for natural resource governance. Standard errors are clustered at the village level. To test hypothesis one, we estimate equations of the form

$$y_{iv} = \beta_1 \tau_i + \beta_2 X_i + \gamma_v + \epsilon_i$$

where y denotes the outcome variable, τ_i denotes the treatment status, X_i is a vector of controls, γ denotes village-level fixed effects, i indexes individual observations and v indexes villages. Here, β_1 captures the coefficient of interest: the effect of assignment into the treatment vignettes on the outcome variables. To test hypotheses two and three, we will estimate equations of the form

$$y_{iv} = \beta_1 \tau_i + \beta_2 M_i + \beta_3 \tau_i \cdot M_i + \beta_4 X_i + \gamma_v + \epsilon_i$$

where M represents the variable by which we hypothesized to find heterogeneous preferences for natural resource governance and all other terms remain the same.¹⁸ In this case, our outcomes of interest are β_1 , the coefficient on the treatment indicator, and β_3 , the coefficient on the interaction variable. For ease of interpretation, figures 3 and 4 in the body of the paper show the marginal effects of treatment by the interacting variables; appendix A.1 shows the full regression tables. Appendix figures A1 and A2 replicate our interactions using the binning estimator from Hainmueller, Mummolo, and Xu (2019), with substantively identical results.

Ultimately, our survey design permits us to observe only some of the underlying variation that distinguishes respondents; more variation likely remains unobserved. Following Oster (2019), we conduct additional sensitivity analysis to calculate the proportion of unobserved variation relative to observed variation which would be necessary to render our results statistically insignificant. This measure, conventionally referred to as δ , captures

18. Our village-level fixed effects absorb the un-interacted effects of M_i where M does not vary within villages.

how robust these results are to unobserved variation which could otherwise render them null. Tables 2 and 3 contain these measures for each outcome variable, for both treatment indicators (the community-led treatment and the state-led treatment). These values suggest that the unobserved variation would have to be at least as large as the observed variation to render our results null.¹⁹

4 Results

Table 2 shows the effect of being treated with the descriptions of natural resource governance on the four binary outcome measures. A positive coefficient would imply greater perceived efficacy for the hypothetical governance regime. Table 2 shows results which are both clear and consistent. Contra H.1a, we see minimal responses to the state-led treatment. Respondents think that a state-led natural resource governance regime will be 0.09 standard deviations more equitable and 0.11 standard deviations more likely to benefit all members of the community. However, respondents did not believe that a state-led system would be more resilient or would reduce conflicts.

In contrast, table 2 provides strong evidence that respondents prefer natural resource governance in which local institutions play a strong role, compared to either the status quo or state-led natural resource governance. Respondents think that a community-led natural resource governance regime would be superior across all measures. Respondents were 0.4 standard deviations more likely to agree that “there would be an increase in resilience to climate-related challenges” (an increase of approximately 25 percent relative to the mean); 0.52 standard deviations more likely to agree that “the system will reduce conflicts over natural resources” (an increase of 28 percent relative to the mean); 0.48 standard deviations more likely to agree that “the resolution of conflicts/disputes will be equitable?” (an increase of 25 percent relative to the mean); and 0.46 standard deviations more likely to agree that “the authorities will voice concerns and make decisions that

19. Demand effects also pose a potential threat to inference: respondents who participate in JASS may feel pressured to give more positive answers or feel that responding positively to the vignettes is the “correct” answer. However, table B1 shows that JASS participation itself does not predict the outcomes, which is evidence against JASS inducing demand effects.

Table 2. Respondents prefer local natural resource governance over the status quo

	Resilience		Reduce conflicts		Equitable		Benefit all members	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Local treatment	0.182*** (0.023)	0.186*** (0.023)	0.227*** (0.022)	0.229*** (0.022)	0.198*** (0.019)	0.200*** (0.019)	0.187*** (0.021)	0.189*** (0.021)
State-led treatment	0.019 (0.018)	0.015 (0.019)	0.038* (0.018)	0.036* (0.018)	0.037* (0.016)	0.034* (0.016)	0.046** (0.017)	0.043** (0.016)
Demographic controls		X		X		X		X
Village FEs	X	X	X	X	X	X	X	X
Mean of outcome	0.726	0.726	0.764	0.764	0.795	0.795	0.8	0.8
δ (local)	1.336	1.806	2.355	2.81	1.861	2.245	1.833	2.233
δ (state)	0.672	0.199	0.798	0.548	1.133	0.645	1.638	0.975
Num.Obs.	3310	3310	3378	3378	3393	3393	3243	3243
R ²	0.129	0.154	0.171	0.184	0.156	0.178	0.171	0.201

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges?’ (columns 1-2), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (columns 3-4), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (columns 5-6), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (columns 7-8). The independent variable is which vignette treatment respondents received. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

benefit all members of the community” (an increase of 23 percent relative to the means). These effects are both statistically and substantively significant. The δ values are well above the conventional threshold of one, suggesting that the results are robust to the presence of unobserved error.

Table 3 shows similar results to table 2, but uses the likert outcome.²⁰ Respondents who received the local governance vignette had 0.39 standard deviations more trust that “land related issues will be managed fairly and transparently.” This effect represents an 13 percent increase over the baseline level of 4.016. In contrast, presenting respondents with the state-led treatment had no effect on trust in the land governance system, relative to the status quo. Across all five outcomes, respondents clearly expressed greater confidence in a community land administration, relative to both existing land administration and a hypothetical state-managed system of land management. Respondents believe that a community-led natural resource governance system would be more resilient, more equitable, and would reduce more conflict.

20. Table A12 in the appendix shows similar results analyzing this outcome variable using an ordinal logit.

Table 3. Respondents prefer local natural resource governance over the status quo

	(1)	(2)
Local treatment	0.548*** (0.063)	0.551*** (0.064)
State-led treatment	0.040 (0.056)	0.040 (0.054)
Demographic controls		X
Village FEs	X	X
Mean of outcome	4.007	4.007
δ (local)	1.523	1.748
δ (state)	2.825	0.819
Num.Obs.	3489	3489
R ²	0.155	0.169

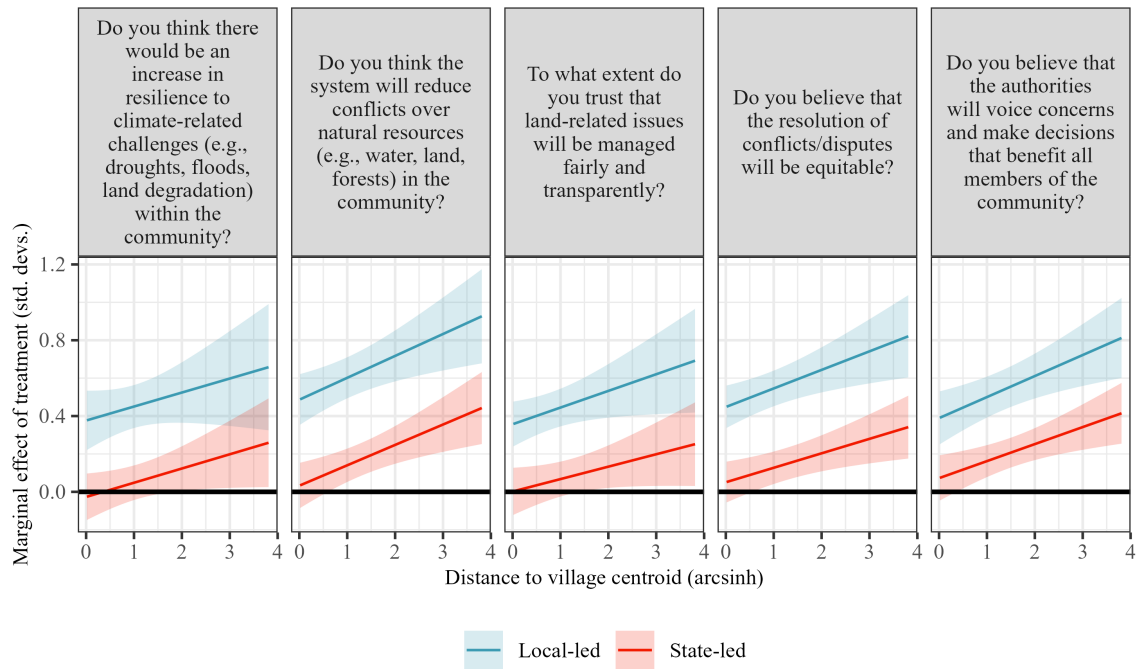
Note: The dependent variable in this model is ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ The independent variable is which vignette treatment respondents received. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

One concern is that our control condition may have different meanings among different respondents because we reference a status quo that likely differs between contexts. These heterogeneous treatment effects may bias any results towards zero by introducing noise into our explanatory variables (Angrist and Pischke 2009). In appendix C we show that these results are reasonably stable across respondents’ pre-treatment preferences for conflict resolution forums. Treatment effects are somewhat higher among respondents who mention using government actors and informal mediators, which could suggest lower levels of confidence. However, appendix C shows that results are homogeneous among respondents who did and did not select the two most common institutions: village conflict resolution committees and community elders.

Beyond this confidence, however, we are also interested in the relationship between other characteristics and respondents’ preferences for natural resource governance. Hy-

Figure 3. Heterogenous treatment effects by household distance to village centroid



This figure shows the marginal effect of treatment with the state-led and community-led natural resource governance vignettes, expressed as standard deviations of each outcome variable, broken out by distance to the village centroid. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level.

pothesis H2.b suggests that allochthones, as opposed to autochthones, or the ‘sons of the soil,’ are more likely to distrust a locally managed natural resource governance system because of a history of discrimination.

Figure 3 shows the marginal effect of being treated by one of the two alternative vignettes by respondent’s distance to the village centroid.²¹ Because settlement patterns generally lead to autochthones occupying the center of the village and allochthones occupying the outskirts, respondents with high distance to the village centroid are more likely to be autochthones. Consequently, H2.A predicts the marginal effect of being treated with the ‘state-led’ vignette to increase with distance to the village centroid. Similarly,

21. Table A1 in the appendix shows full regression tables.

if H2.B is correct, then we would expect the marginal effect of being treated with the ‘local-led’ vignette to decrease with the respondent’s distance to the village centroid.

In line with H.2A, figure 3 shows that support for the state-led treatment does indeed increase with distance to the village centroid. Among respondents who were closest to the village centroid, the marginal effect of being treated with the ‘state-led’ vignette was statistically indistinguishable from zero. In contrast, for respondents furthest from the village centroid (those most likely to be allochthones), the marginal effect of being treated with the state-led treatment was an increase of 0.25 to 0.44 standard deviations. Allochthones trust the state-led system more than autochthones.

However, contra H2.B, figure 3 also shows that the marginal effects of being treated with the ‘local-led’ vignette increase with distance to the village centroid. In other words, allochthones express greater support for a hypothetical community led natural resource governance system. Even among respondents who were furthest from the village centroid, respondents expressed greater confidence in the locally led natural resource governance system than in the state-led natural resource governance system. One potential explanation is that these slopes merely reflect allochthones deep-seated distrust of existing systems. However, table A1 shows that distance to the village centroid by itself is not a statistically significant predictor of these outcomes. In other words, autochthones and allochthones express similar levels of trust in the status quo of natural resource governance.²²

These results are puzzling: they suggest that allochthones trust a hypothetical community based natural resource governance regime even more than autochthones. Previous research on the political economy of natural resource governance suggests the opposite: members of an out-group would have less confidence in regimes vulnerable to capture by local elites (Ribar 2025; Boone 2014). H.2b suggested that allochthones, when presented with the idea of ‘community-led natural resource governance,’ believe that it

22. One alternative explanation here is that respondents who live closer to the village centroid are simply better informed about village politics. However, among respondents who are above the median distance to the centroid, 58.4 percent feel that they “participate in making decisions in terms of access, use and management of resources and related disputes in [their] community;” among those below the median distance to the village centroid, 48 percent feel that they participate. These data suggest that participation does not meaningfully decline with distance—if anything, the opposite.

will be captured by elites, and thus discriminatory towards allochthones as the local out-group.

What explains these unexpected results? When we presented respondents with the community-based natural resource governance regime vignette, H2.B implied that allochthone respondents would not believe that such a regime could be inclusive. Does this assumption hold? One answer to this question comes from the JASS intervention itself, which could function as a proof-of-concept for allochthones when it comes to inclusive natural resource governance.²³ Allochthones who participated in JASS may be more likely to believe that a community-led natural resource governance regime could be genuinely inclusive. Table B1 shows that the marginal effects of being treated with the community-led vignette are somewhat lower among allochthones who participated in JASS. If JASS participation contributes to respondents' perceptions of community-led natural resource governance, then respondents may be cautiously optimistic about the possibilities for inclusive community-led natural resource governance.

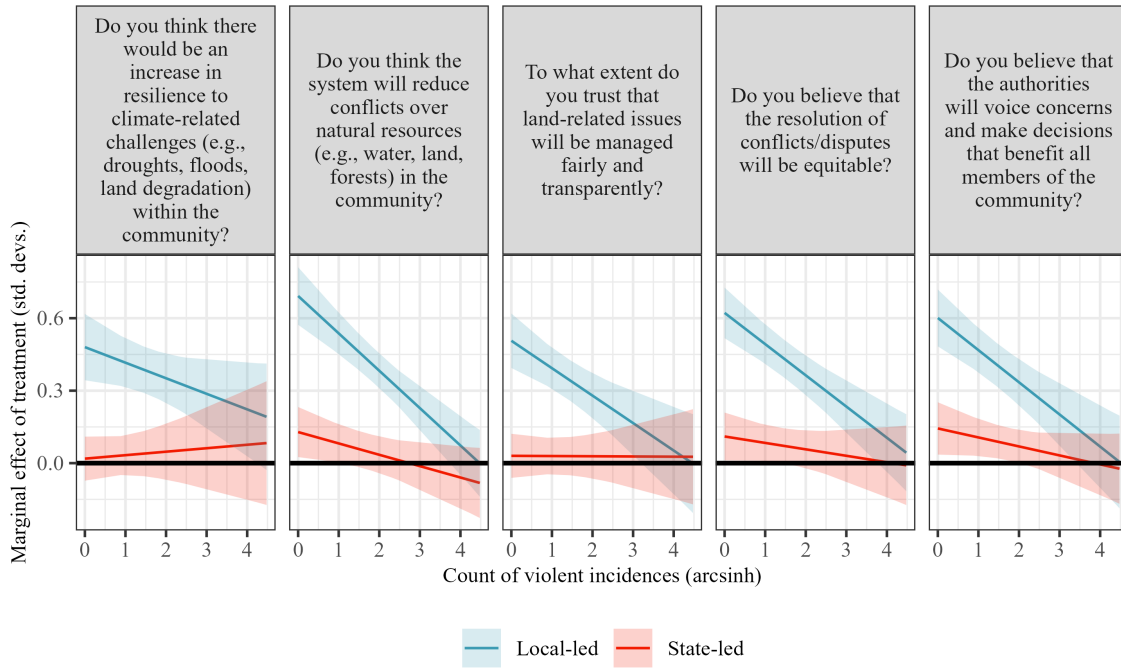
Several of our outcome variables address inclusivity directly. If respondents equated community-led natural resource governance with capture by local elites and subsequent discrimination, then we would expect to see completely null results when asking allochthones (1) 'To what extent do you trust that land-related issues will be managed fairly and transparently?' (2) 'Do you believe that the resolution of conflicts/disputes will be equitable?' and (3) 'Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?' Contrary to our expectations, allochthones believed that a community-led natural resource governance regime could in fact be inclusive.²⁴ Respondents do believe in the potential for inclusive natural resource governance, even if this belief remains guarded. In other words, allochthones are evaluating the proposed natural resource governance institutions on their own terms, rather than in reference to prior experiences.

Finally, we are also interested in how exposure to violence affects confidence in these

23. These results are not pre-registered, and should be treated as exploratory rather than confirmatory.

24. Alternatively, allochthone respondents could think that norms of formal property rights will nevertheless pervade community-led natural resource governance. Aldashev et al. (2012, 798) refer to this mechanism as a magnet effect: "the formal law can actually pull custom in its direction, thereby causing a progressive evolution of the prevailing mores."

Figure 4. Heterogeneous treatment effects by exposure to ACLED events



This figure shows the marginal effect of treatment with the state-led and community-led natural resource governance vignettes, expressed as standard deviations of each outcome variable, and broken out by exposure to violent events status. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level.

proposed natural resource governance regimes. Figure 4 shows the marginal effects of being treated with the state-led and local-led vignettes by exposure to violent incidents. If H_3 were correct, we would expect to see the marginal effects of being treated with the ‘state-led’ vignette to be lesser among respondents who had greater exposure to violence. Figure 4 does indeed provide support for this hypothesis. However, the marginal effect of the ‘state-led’ treatment is not statistically significant in much of figure 4, suggesting that $H_{3.A}$ encounters floor effects. There may be a negative relationship between confidence in state-led natural resource governance and exposure to conflict but it is difficult to detect because confidence in state-led natural resource governance is quite low.

Figure 4 also reveals a robust finding that we did not hypothesize: exposure to vi-

olence *also* decreases confidence in locally led natural resource governance. Where respondents have been exposed to zero ACLED events, the marginal effect of being treated with the local-led vignette is a 0.46 to 0.60 standard deviation increase in respondents' confidence in the hypothetical regime. Where exposure to violence is high, the marginal effect of being exposed to the local-led vignette is statistically distinguishable from zero. In other words, being exposed to violence decreases confidence in all hypothetical natural resource governance arrangements, not just state-led natural resource governance.²⁵ At the highest levels of exposure to violence, respondents did not believe that either proposed natural resource governance regime would be an improvement relative to the status quo. While exposure to conflict may decrease respondents' beliefs in the state's ability to enforce property rights, this loss of trust also seems to spill over to the community's ability to enforce property rights.

5 Discussion and Conclusion

Low state capacity, security challenges, and shrinking resource pools have made natural resource governance difficult in countries like Mali and Niger. In response, both donors and African governments have attempted to expand the provision of natural resource governance, either by providing formal property rights (Deininger and Goyal 2024) or by standing up community-led conflict resolution institutions (Christensen et al. 2025; Hartman, Blair, and Blattman 2021; Ribar et al. 2025). The community-led interventions in particular are predicated on the idea that local and inclusive governance fosters higher levels of trust, legitimacy, and cooperation (Ostrom 2015). For natural resource governance to be successful, constituents must have confidence in the system; where constituents have multiple options for conflict resolution institutions, they need not rely on natural resource governance that they do not feel would be effective. Within this challenging environment, what kinds of institutions do citizens believe could improve natural resource management, resolve conflicts, and boost climate resilience?

25. In our current specifications, we cannot show the relationship between exposure to violence and the control condition of the vignette experiment, because the village fixed effects absorb the un-interacted effect of violence.

This paper provides a three part answer to this question. First, respondents overwhelmingly prefer a community-led natural resource governance relative to either a state-led natural resource governance regime or the status quo. These results are stark: relative to the status quo, being presented with the community-led treatment increased confidence in the proposed system by 0.4 to 0.52 standard deviations—increases of 23 percent to 28 percent over the mean. These results align with a growing movement within natural resource governance policy to empower communities to monitor their own property rights and adjudicate their own disputes (Callen, Weigel, and Yuchtman 2024; Deininger and Goyal 2024).

Second, these preferences are similar among both local in-groups (autochthones) and local out-groups (allochthones). Existing literature suggests that households' position within local hierarchies will affect their confidence in the various institutions to which they could bring a conflict, thereby affecting their choice of forum (Acemoglu et al. 2020; Funjika and Honig 2024; Winters and Conroy-Krutz 2021). Respondents located further from the village centroid, i.e. those more likely to belong to the out-group, have increased confidence in both the community-led and state-led natural resource governance treatment vignettes, relative to the status quo. In other words—contrary to both our own hypotheses and the existing literature—respondents who belong to local outgroups believe that a community-led natural resource governance regime could be both inclusive and effective.

Third, exposure to violence decreases confidence in both community-led and state-led natural resource governance regimes. These findings align with our theoretical expectations that exposure to violence signals the state's inability to enforce property rights. Violent incidents show respondents that the state lacks a monopoly on violence. Where access to natural resources forms the core of many conflicts (Benjaminsen and Ba 2024; Raleigh, Nsaibia, and Dowd 2021; Hansen 2024), exposure to violence directly indicates that the state has failed to resolve natural resource conflicts in the past. These findings challenge existing literature which identifies sequences for peacebuilding (Paris 2004; Besley and Persson 2010), as well as literature on using community-based institutions for peacebuilding. If exposure to violence reduces confidence in both state-led and community-led institutions, then pivots to community-led institutions as a first-step

for peacebuilding are unlikely to be successful. This article speaks to a growing literature which posits that that community institutions can substitute for state-led institutions (Van Der Windt et al. 2019; Balán et al. 2022; Greico 2025) by identifying how ongoing violence undermines confidence in such institutions. Institutional pluralism, therefore, may not be a buffer against the negative consequences of eroded state capacity: having access to multiple forums is no consolation when rural citizens do not trust any of the available forums.

Our survey is embedded within a capacity building, resilience, and natural resource management intervention meant to address stress factors that may contribute to conflict. Consequently, the villages we survey are located in climate-stressed and rural areas with minimal state presence, where subsistence agriculture and livestock raising predominate. Our sample represents these implementation geographies rather than the national populations of Mali or Niger. However, such conditions are not unique to Mali or Niger. Climate stress, low state capacity, and lingering conflicts over natural resources are common conditions across both the Sahel and sub-Saharan Africa more broadly (McGuirk and Nunn 2025). Given the frequency of such settings, constraining our external validity to similarly rural or fragile contexts does not seem overly limiting.

Future research could better unpack the specific elements of community-based natural resource governance which condition perceived equity and efficiency and make sense of these unexpected results. Varying the features of community-based natural resource governance could illustrate why and when members of local out groups would perceive community land governance to be equitable. More specifically, by decomposing the bundled treatments in our vignettes, future research could isolate the specific mechanisms which drive support for and confidence in natural resource governance regimes. While we measure the perceived effectiveness of alternative natural resource regimes, other choice-based designs could also measure preferences more directly.

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Author contributions

Alma Bezares Calderón: Conceptualization, Investigation, writing – review & editing. Simon Mercer: Conceptualization, Funding acquisition. Ifeoluwa M. Olawole: Conceptualization, Investigation, writing – review & editing. Matthew K. Ribar: Data curation, Investigation, Methodology, Visualization, Writing - original draft, writing – review & editing.

Conflict of interest

The authors do not report any conflict of interest in the writing and publication of this article.

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Data availability

The dataset, codebook, and R scripts for the empirical analysis in this article, along with the Online Appendix, are available at <https://www.prio.org/journals/jpr/replicationdata>. All analyses were conducted using R version 4.5.0.

Declaration of AI usage

The authors have not used any Generative or Assistive AI tools in the writing of this article.

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Online Appendix:
Belonging, violence, and natural resource governance: experimental
evidence from Mali and Niger

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A Additional specifications

A.1 Tables for marginal effects plots

Table [A1](#) shows the regression coefficients which generate figure [3](#). The table shows the direct effect of autochthony (measured via distance to the village centroid) on our five outcome variables. Distance by itself does not affect most of the outcomes, with the exception of responses to ‘to what extent do you trust that land-related issues will be managed fairly and transparently?’ (column three). These results suggest that allochthony is not associated with different responses under the control condition (the status quo). However, the fact that allochthones think that the status quo is more fair and transparent is also counter to our expectations. Next, table [A2](#) shows the regressions we use for figure [4](#).

Table A1. Households further from village centroids are more responsive to treatment

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.168*** (0.036)	0.206*** (0.029)	0.500*** (0.085)	0.181*** (0.023)	0.156*** (0.029)
State-led treatment	-0.012 (0.028)	0.014 (0.026)	0.002 (0.089)	0.020 (0.022)	0.029 (0.024)
Distance	-0.023 (0.035)	-0.012 (0.034)	0.201* (0.090)	0.024 (0.031)	0.018 (0.032)
Local * Distance	0.033 (0.025)	0.049** (0.017)	0.123* (0.061)	0.040** (0.014)	0.044** (0.015)
State * Distance	0.034 (0.018)	0.046** (0.014)	0.092 (0.054)	0.031** (0.011)	0.036** (0.012)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.726	0.764	4.007	0.795	0.8
Num.Obs.	2837	2872	2934	2875	2762
R2	0.168	0.200	0.190	0.192	0.220

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges’ (column 1), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (column 2), ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (column 3), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (column 4), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (column 5). The independent variable is which vignette treatment respondents received. This table measures allochthony using the distance to the village centroid (in meters, using the inverse hyperbolic sine transformation to normalize). Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table A2. Exposure to ACLED events is associated with a smaller effect for the ‘local governance’ treatment

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.214*** (0.031)	0.294*** (0.026)	0.709*** (0.080)	0.251*** (0.022)	0.240*** (0.024)
State-led treatment	0.008 (0.021)	0.055* (0.022)	0.042 (0.065)	0.045* (0.020)	0.057* (0.022)
Local * Violent events	−0.029 (0.015)	−0.066*** (0.010)	−0.159*** (0.042)	−0.052*** (0.010)	−0.053*** (0.011)
State * Violent events	0.006 (0.015)	−0.020* (0.010)	−0.001 (0.037)	−0.011 (0.010)	−0.015 (0.010)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.726	0.764	4.007	0.795	0.8
Num.Obs.	3301	3369	3478	3384	3234
R ²	0.156	0.191	0.173	0.182	0.206

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges’ (column 1), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (column 2), ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (column 3), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (column 4), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (column 5). The independent variable is the treatment vignette that respondents received. This table counts violent incidents as the sum of ACLED events in the year before the survey within 25 kilometers of the village centroid, regularized using the inverse hyperbolic sine transformation. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects and survey weights. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

A.2 All measures of conflict exposure

Table A3 shows the results of interacting our two treatment indicators with the self reported exposure to violence. Treatment effects are homogeneous across levels of self-reported exposure to violence. However, it makes sense that reporting the activities of armed groups would be subject to social desirability bias: respondents may have been concerned that reporting such activities would bring state reprisals, which they would wish to avoid. Consequently, we prefer the ACLED measures of exposure to conflict.

Table A3. Self-reported exposure to violence is not associated with heterogenous treatment effects

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.207*** (0.020)	0.233*** (0.022)	0.573*** (0.068)	0.207*** (0.020)	0.201*** (0.023)
State-led treatment	0.034 (0.019)	0.030 (0.019)	0.029 (0.066)	0.034 (0.019)	0.044** (0.017)
Local * Violence	-0.047 (0.041)	-0.026 (0.049)	-0.145 (0.165)	-0.047 (0.041)	-0.075 (0.044)
State * Violence	-0.004 (0.051)	0.034 (0.056)	0.072 (0.171)	-0.004 (0.051)	-0.010 (0.044)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.795	0.764	4.007	0.795	0.8
Num.Obs.	3393	3378	3489	3393	3243
R2	0.179	0.185	0.170	0.179	0.203

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges’ (column 1), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (column 2), ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (column 3), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (column 4), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (column 5). The independent variable is the treatment vignette that respondents received. Exposure to violence is measured through responses to ‘To your knowledge, have there been any violent confrontations or conflicts in your community in the last six months?’ Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects and survey weights. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

In our pre-analysis plan, we did not specify a specific strategy to aggregate ACLED events. However, this section shows that our results are not reliant on a specific formulation of ACLED event counts. Each table shows results for one of the five outcome variables. ACLED event counts vary by (1) the period of time in which we count ACLED

Table A4. Coefficients are stable across ACLED specifications for responses to ‘Do you think there would be an increase in resilience to climate-related challenges within the community?’

Period	Radius	Violence * Local			Violence * State		
		Coef.	Std.Err.	T-score	Coef.	Std.Err.	T-score
No transformation							
Three years	10 Km.	-0.002	0.001	-4.544	0.000	0.001	0.415
Two years	10 Km.	-0.003	0.001	-4.359	0.000	0.001	0.463
One year	10 Km.	-0.009	0.002	-4.307	0.000	0.002	-0.149
Three years	25 Km.	-0.001	0.000	-2.487	0.000	0.000	0.983
Two years	25 Km.	-0.001	0.001	-2.246	0.001	0.001	1.025
One year	25 Km.	-0.004	0.001	-2.465	0.002	0.002	0.939
Inverse hyperbolic sine transformation							
Three years	10 Km.	-0.029	0.013	-2.187	0.007	0.014	0.475
Two years	10 Km.	-0.031	0.014	-2.121	0.008	0.016	0.512
One year	10 Km.	-0.045	0.017	-2.618	0.003	0.018	0.188
Three years	25 Km.	-0.023	0.012	-1.881	0.005	0.011	0.471
Two years	25 Km.	-0.025	0.013	-1.984	0.005	0.012	0.451
One year	25 Km.	-0.029	0.015	-1.906	0.006	0.015	0.439

Note: This table replicates the coefficients on the interaction effects from table A2 using a variety of different specifications ACLED data.

events; (2) the size of the radius around the village in which we include events; and (3) whether we use an inverse hyperbolic sine transformation. The conclusion we can draw from these tables is that our results do not depend on a specific specification of ACLED events.

Table A5. Coefficients are stable across ACLED specifications for responses to ‘Do you think the system will reduce conflicts over natural resources in the community?’

Period	Radius	Violence * Local			Violence * State		
		Coef.	Std.Err.	T-score	Coef.	Std.Err.	T-score
No transformation							
Three years	10 Km.	-0.004	0.001	-5.304	-0.001	0.000	-2.230
Two years	10 Km.	-0.005	0.001	-5.195	-0.001	0.001	-2.258
One year	10 Km.	-0.013	0.003	-4.896	-0.003	0.001	-2.168
Three years	25 Km.	-0.002	0.000	-8.982	0.000	0.000	-2.460
Two years	25 Km.	-0.003	0.000	-9.069	-0.001	0.000	-2.413
One year	25 Km.	-0.007	0.001	-8.685	-0.002	0.001	-2.442
Inverse hyperbolic sine transformation							
Three years	10 Km.	-0.054	0.010	-5.297	-0.016	0.009	-1.730
Two years	10 Km.	-0.057	0.011	-5.220	-0.018	0.010	-1.852
One year	10 Km.	-0.069	0.014	-4.912	-0.022	0.012	-1.819
Three years	25 Km.	-0.056	0.009	-6.538	-0.020	0.008	-2.493
Two years	25 Km.	-0.058	0.009	-6.699	-0.019	0.008	-2.280
One year	25 Km.	-0.066	0.010	-6.365	-0.020	0.010	-2.098

Note: This table replicates the coefficients on the interaction effects from table A2 using a variety of different specifications ACLED data.

Table A6. Coefficients are stable across ACLED specifications for responses to ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’

Period	Radius	Violence * Local			Violence * State		
		Coef.	Std.Err.	T-score	Coef.	Std.Err.	T-score
No transformation							
Three years	10 Km.	-0.011	0.004	-2.503	-0.003	0.004	-0.936
Two years	10 Km.	-0.016	0.007	-2.467	-0.005	0.005	-0.927
One year	10 Km.	-0.037	0.016	-2.353	-0.014	0.014	-0.959
Three years	25 Km.	-0.006	0.001	-4.261	-0.001	0.001	-0.720
Two years	25 Km.	-0.009	0.002	-4.635	-0.001	0.002	-0.629
One year	25 Km.	-0.023	0.005	-4.314	-0.003	0.004	-0.664
Inverse hyperbolic sine transformation							
Three years	10 Km.	-0.162	0.041	-4.005	-0.038	0.038	-0.999
Two years	10 Km.	-0.177	0.045	-3.950	-0.047	0.042	-1.128
One year	10 Km.	-0.220	0.061	-3.621	-0.074	0.058	-1.289
Three years	25 Km.	-0.127	0.031	-4.037	-0.008	0.029	-0.280
Two years	25 Km.	-0.136	0.034	-3.964	-0.002	0.030	-0.069
One year	25 Km.	-0.159	0.042	-3.799	-0.001	0.037	-0.033

Note: This table replicates the coefficients on the interaction effects from table A2 using a variety of different specifications ACLED data.

Table A7. Coefficients are stable across ACLED specifications for responses to ‘Do you believe that the resolution of conflicts/disputes will be equitable?’

Period	Radius	Violence * Local			Violence * State		
		Coef.	Std.Err.	T-score	Coef.	Std.Err.	T-score
No transformation							
Three years	10 Km.	-0.003	0.001	-5.216	0.000	0.000	-1.335
Two years	10 Km.	-0.005	0.001	-5.089	-0.001	0.000	-1.314
One year	10 Km.	-0.011	0.002	-4.598	-0.002	0.001	-1.947
Three years	25 Km.	-0.002	0.000	-9.248	0.000	0.000	-0.504
Two years	25 Km.	-0.002	0.000	-9.237	0.000	0.000	-0.411
One year	25 Km.	-0.006	0.001	-9.112	-0.001	0.001	-0.604
Inverse hyperbolic sine transformation							
Three years	10 Km.	-0.045	0.010	-4.480	-0.006	0.009	-0.681
Two years	10 Km.	-0.048	0.011	-4.539	-0.008	0.010	-0.817
One year	10 Km.	-0.063	0.014	-4.560	-0.014	0.012	-1.156
Three years	25 Km.	-0.044	0.008	-5.522	-0.009	0.008	-1.109
Two years	25 Km.	-0.046	0.008	-5.665	-0.009	0.008	-1.038
One year	25 Km.	-0.052	0.010	-5.447	-0.011	0.010	-1.090

Note: This table replicates the coefficients on the interaction effects from table A2 using a variety of different specifications ACLED data.

Table A8. Coefficients are stable across ACLED specifications for responses to ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’

Period	Radius	Violence * Local			Violence * State		
		Coef.	Std.Err.	T-score	Coef.	Std.Err.	T-score
No transformation							
Three years	10 Km.	-0.003	0.001	-5.027	-0.001	0.000	-2.117
Two years	10 Km.	-0.004	0.001	-4.911	-0.001	0.000	-2.113
One year	10 Km.	-0.011	0.003	-4.225	-0.003	0.001	-2.510
Three years	25 Km.	-0.001	0.000	-5.983	0.000	0.000	-1.283
Two years	25 Km.	-0.002	0.000	-5.491	0.000	0.000	-1.156
One year	25 Km.	-0.005	0.001	-6.007	-0.001	0.001	-1.273
Inverse hyperbolic sine transformation							
Three years	10 Km.	-0.045	0.010	-4.384	-0.010	0.009	-1.095
Two years	10 Km.	-0.049	0.011	-4.555	-0.011	0.009	-1.215
One year	10 Km.	-0.066	0.014	-4.786	-0.016	0.011	-1.519
Three years	25 Km.	-0.044	0.009	-4.827	-0.017	0.008	-2.186
Two years	25 Km.	-0.047	0.010	-4.944	-0.015	0.008	-1.827
One year	25 Km.	-0.053	0.011	-4.751	-0.015	0.010	-1.530

Note: This table replicates the coefficients on the interaction effects from table A2 using a variety of different specifications ACLED data.

A.3 Alternative measures of autochthony

This section shows the results of estimating the primary interactions using our pre-registered measurements of autochthony. Table A9 uses whether a respondent self-identifies as a ‘host’ in their community; table A10 uses whether a respondent self-reports as being ethnically Tuareg or Peulh/Fulani. However, both measures leave us severely underpowered to detect our treatment effects, as only 69 respondents reported being ‘hosts’ and only 282 reported being Tuaregs or Peulh/Fulani. Consequently, we prefer the distance to centroid measure we deploy in the body of the paper.

Table A9. Self-reported autochthones prefer local governance, but power is too low

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.177*** (0.023)	0.222*** (0.023)	0.526*** (0.065)	0.191*** (0.019)	0.181*** (0.021)
State-led treatment	0.009 (0.018)	0.032 (0.017)	0.037 (0.054)	0.030 (0.015)	0.039* (0.016)
Local * Allochthone	0.514*** (0.121)	0.428*** (0.115)	1.360*** (0.293)	0.513*** (0.114)	0.448*** (0.116)
State * Allochthone	0.235 (0.143)	0.183 (0.160)	−0.028 (0.337)	0.178 (0.154)	0.142 (0.132)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.726	0.764	4.007	0.795	0.8
Num.Obs.	3310	3378	3489	3393	3243
R ²	0.170	0.192	0.176	0.190	0.212

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges’ (column 1), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (column 2), ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (column 3), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (column 4), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (column 5). The independent variable is which vignette treatment respondents received. Allochthony is self-reported using by asking if the respondent is a host/autochthone or if they have moved, returned, or are a refugee. 70 respondents identified as allochthones; 3,617 did not. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Alternatively, one could simply measure whether a respondent is an allochthone based on whether they are the ethnic majority within their own village. There are two problems with this approach. First, this survey presents only a noisy version of a village’s demography. It may be the case that the majority ethnicity among respondents is not

Table A10. No heterogenous effects among herding/allochthonous ethnicities

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.190*** (0.024)	0.237*** (0.023)	0.560*** (0.068)	0.203*** (0.019)	0.194*** (0.021)
State-led treatment	0.015 (0.020)	0.043* (0.018)	0.051 (0.059)	0.035* (0.016)	0.042* (0.017)
Local * Allochthone	−0.049 (0.070)	−0.094 (0.068)	−0.108 (0.126)	−0.039 (0.052)	−0.060 (0.068)
State * Allochthone	−0.003 (0.060)	−0.095 (0.050)	−0.140 (0.162)	−0.017 (0.042)	0.010 (0.034)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.726	0.764	4.007	0.795	0.8
Num.Obs.	3310	3378	3489	3393	3243
R2	0.154	0.185	0.170	0.178	0.201

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges’ (column 1), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (column 2), ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (column 3), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (column 4), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (column 5). The independent variable is which vignette treatment respondents received. This table measures allochthony using the herding ethnicities (Tuareg and Peulh); 298 respondents reported belonging to these ethnicities and 3398 did not. 70 respondents identified as allochthones; 3,617 did not. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

the same as the majority ethnicity among villages. Second, a village may be majority allochthone. Autochthones descent from the original lineage of the villages, but in some areas of West Africa, later arrivals begin to outnumber the original lineage. Nevertheless, table A11 shows heterogeneous treatment effects for the two vignettes based on whether the respondent is a member of the ethnic majority within their village. It includes only villages where one ethnicity comprises at least fifty percent of respondents (i.e. an ethnic majority exists) and with at least ten respondents.

Table A11. Treatment effects are lower among local ethnic outsiders

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.193*** (0.028)	0.267*** (0.025)	0.649*** (0.075)	0.233*** (0.020)	0.223*** (0.023)
State-led treatment	-0.011 (0.022)	0.037 (0.022)	0.061 (0.068)	0.032 (0.019)	0.042* (0.020)
Ethnic outsider	0.022 (0.041)	0.067 (0.039)	0.299** (0.102)	0.049 (0.042)	0.073* (0.036)
Local * outsider	-0.010 (0.057)	-0.094 (0.049)	-0.172 (0.157)	-0.071 (0.049)	-0.093 (0.049)
State * outsider	0.070 (0.074)	-0.024 (0.071)	-0.069 (0.195)	-0.011 (0.071)	0.011 (0.052)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.702	0.735	3.921	0.769	0.772
Num.Obs.	2671	2722	2813	2739	2640
R ²	0.145	0.170	0.158	0.164	0.186

Note: The dependent variables in this model are ‘Do you think there would be an increase in resilience to climate-related challenges’ (column 1), ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (column 2), ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (column 3), ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ (column 4), and ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ (column 5). The explanatory variables include the treatment received by the respondent and whether they are a local ethnic outsider (i.e. not the ethnic majority within villages). This table includes only villages with greater than ten respondents and in which one ethnicity comprises at least 50 percent of the respondents. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village type fixed effects and survey weights. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

A.4 Ordinal logits

Table Ar2. Respondents prefer local natural resource governance over the status quo

	(1)	(2)
Local treatment	0.860*** (0.089)	0.875*** (0.089)
State-led treatment	0.059 (0.081)	0.055 (0.081)
Demographic controls		X
Village FEs	X	X
Num.Obs.	3489	3489

Note:

This table replices table 3 using an ordinal logit instead of ordinary least squares regression. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

We use ordinary least squares to estimate our results, but we note that one of the outcomes is a five point Likert-scale, which means an ordinal logit would be, in theory, more suitable. This section shows that replicating our primary results with an ordinal logit for this question yields a substantially identical set of results.

A.5 Hand-coded village centroids

One concern is that noise in the GPS data could contaminate our measurements of the village centroids, making our measurement of household distance to the village centroids an inaccurate measure of the true distance to village centers. To resolve this concern, we hand-coded the village centers, using either central open spaces or intersections as the village centers. Across the 79 villages, the mean difference between the village centroid and the hand-coded village center is 447 meters; the median distance is 164 meters.

Table A13. Subsetting to hand-verified village centroids yields similar results

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.192*** (0.039)	0.220*** (0.032)	0.518*** (0.090)	0.181*** (0.025)	0.150*** (0.032)
State-led treatment	0.001 (0.031)	0.019 (0.029)	−0.001 (0.099)	0.023 (0.024)	0.021 (0.027)
Distance	−0.082 (0.054)	−0.042 (0.050)	0.129 (0.095)	−0.019 (0.042)	−0.023 (0.046)
Local * Distance	0.017 (0.029)	0.043* (0.018)	0.087 (0.063)	0.031** (0.012)	0.037** (0.012)
State * Distance	0.041 (0.027)	0.048* (0.019)	0.124 (0.070)	0.037* (0.015)	0.043** (0.016)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.73	0.768	4.027	0.807	0.811
Num.Obs.	2082	2114	2167	2108	1998
R2	0.158	0.197	0.191	0.185	0.221

Note: This table replicates table A1, but subsetting to villages where the distance between village centroids (used elsewhere in the article) and hand-coded village centers is less than 500 meters. Control variables include age, age squared, sex, and which (if any) JASS activities the respondent participated. All regressions use OLS with village type fixed effects. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

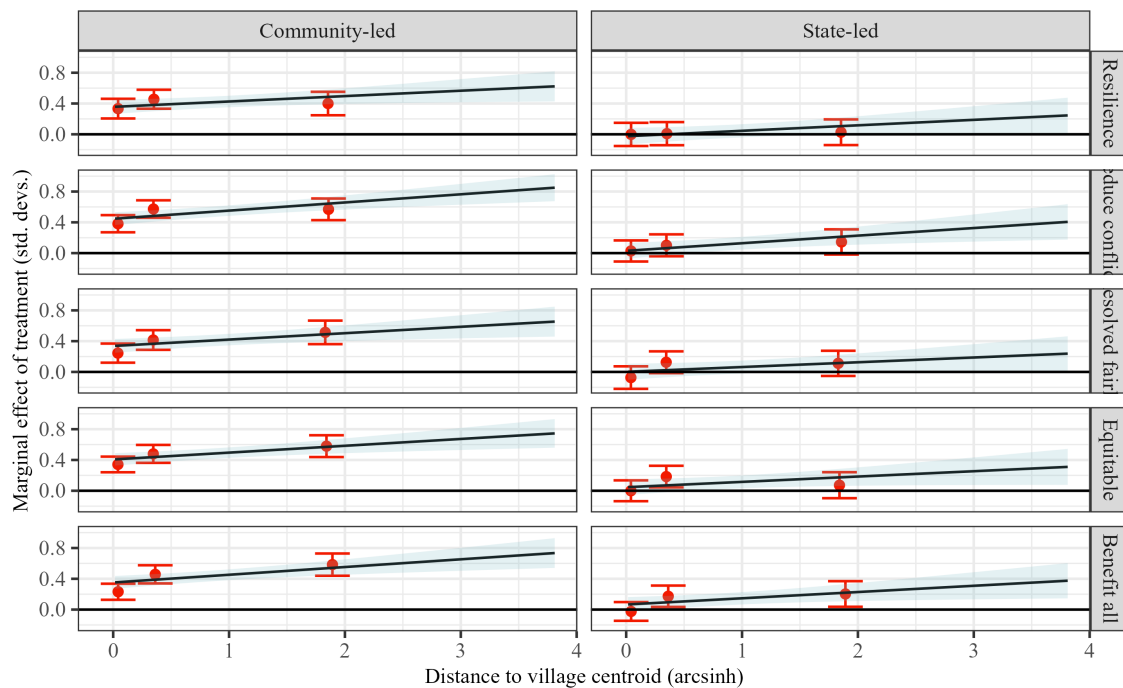
We cannot code households' distance to the newly coded village centers because the country offices shared only anonymized versions of these data. Household GPS coordinates would, of course, de-anonymize these data. However, table A13 replicates our main analyses using distance to the village centroid, but using only households in villages where the distance between the village centroid and the hand-coded village center is less than 500 meters. In other words, this table excludes the villages where this error

may be driving our results. Table A13 shows substantively identical results to our main analyses (accounting for smaller sample sizes), which gives confidence that the distance to village centroids is capturing real village geographies, rather than statistical noise.

A.6 Binning estimates of interactions

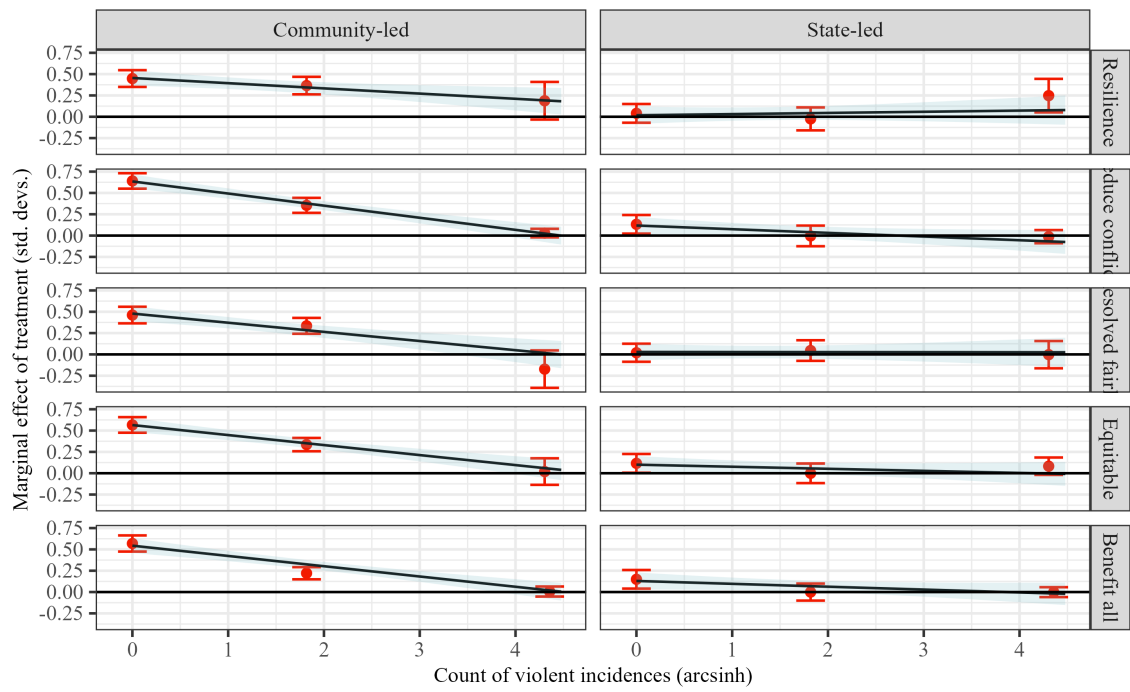
While linear interactive models are relatively simple to understand and interpret, they may conceal non-linear relationships between our treatment effect and the variable with which we interact it: distance to the village centroid for hypotheses H.2a and H.2b, and the count of violent events for hypotheses H.3. In this appendix section, we replicate figures 3 and 4 using the binning estimator from Hainmueller, Mummolo, and Xu (2019). Our results are substantially identical when using this binning estimator.

Figure A1. Heterogeneous treatment effects by autochthony (binning estimate)



This figure replicates figure 3 while using the binning estimator from Hainmueller, Mummolo, and Xu (2019).

Figure A2. Heterogeneous treatment effects by conflict exposure (binning estimate)



This figure replicates figure 4 while using the binning estimator from Hainmueller, Mummolo, and Xu (2019).

B Heterogeneous effects by intervention

One threat to inference is that the JASS intervention itself may have shifted results due to demand effects. In other words, respondents in JASS villages might be aware of the program objectives, and thus give enumerators more positive responses to questions about equitable natural resource governance. If respondents who participated in JASS thought that they ought to give more favorable responses, then we would expect a positive coefficient on JASS participation independent of the treatment condition. Table B1 shows no such coefficient. Moreover, we show only sporadic heterogeneous treatment effects across respondents who did and did not participate in JASS. These results do not support a claim that demand effects from JASS drive our primary results.

Table B1. Heterogeneous effects by JASS participation show no evidence of demand effects

	Resilience	Reduce conflicts	Fairly	Equitable	Benefit all members
	(1)	(2)	(3)	(4)	(5)
Local treatment	0.154*** (0.036)	0.171*** (0.029)	0.463*** (0.102)	0.164*** (0.028)	0.133*** (0.029)
State-led treatment	0.008 (0.035)	0.015 (0.029)	0.038 (0.097)	0.039 (0.025)	0.005 (0.028)
Any JASS participation	-0.015 (0.033)	-0.039 (0.033)	0.016 (0.088)	-0.025 (0.030)	-0.028 (0.025)
Local * JASS	0.045 (0.043)	0.082* (0.037)	0.124 (0.119)	0.051 (0.033)	0.078* (0.034)
State * JASS	0.020 (0.039)	0.036 (0.036)	0.013 (0.130)	0.000 (0.033)	0.062 (0.038)
Demographic controls	X	X	X	X	X
Village FEs	X	X	X	X	X
Mean of outcome	0.726	0.764	4.007	0.795	0.8
Num.Obs.	3310	3378	3489	3393	3243
R ²	0.133	0.175	0.159	0.160	0.177

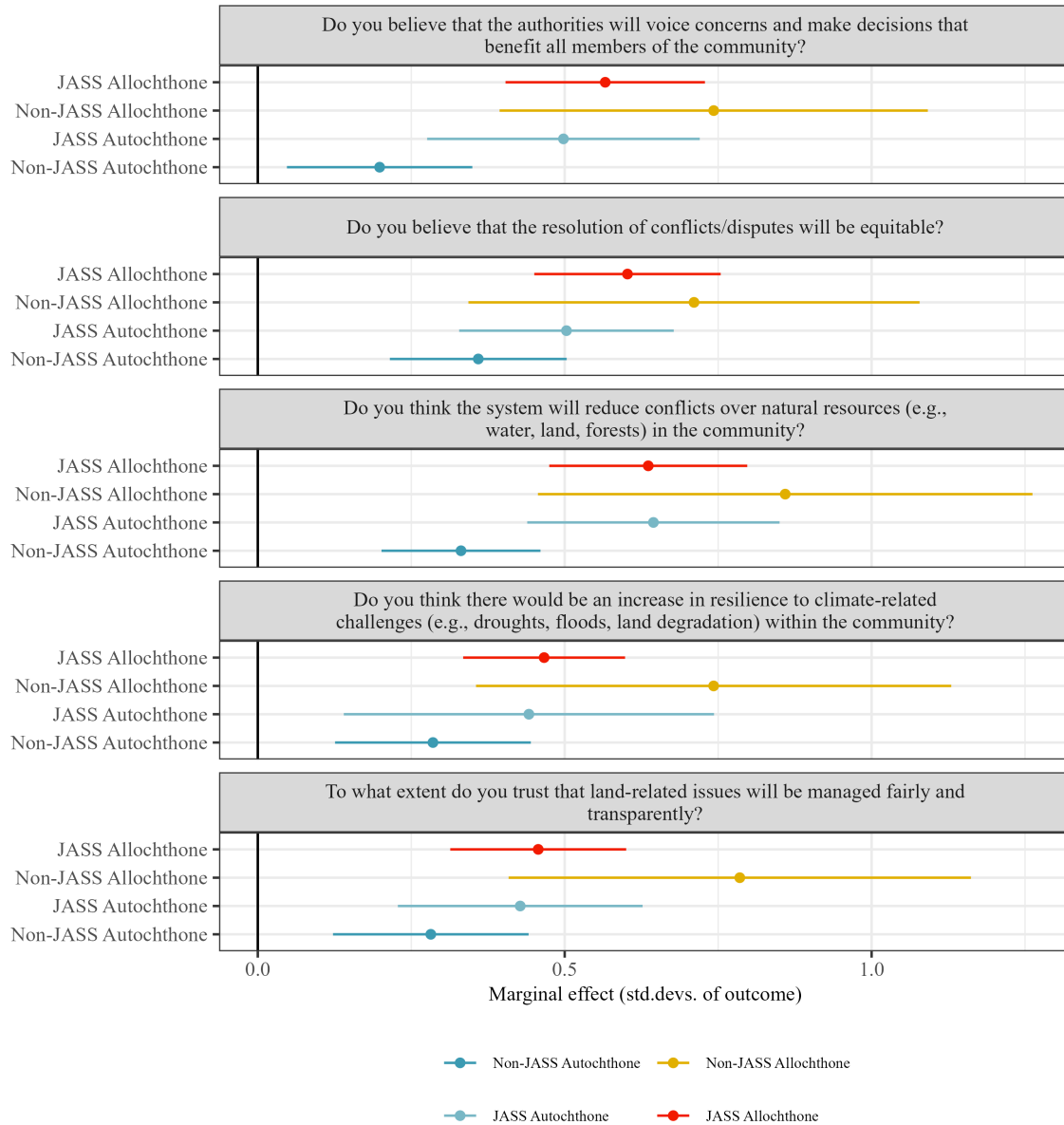
Note: The dependent variables in this model are (1) ‘Do you think there would be an increase in resilience to climate-related challenges?’ (2) ‘Do you think the system will reduce conflicts over natural resources (e.g., water, land, forests) in the community?’ (3) ‘To what extent do you trust that land-related issues will be managed fairly and transparently?’ (4) ‘Do you believe that the resolution of conflicts/disputes will be equitable?’ and (5) ‘Do you believe that the authorities will voice concerns and make decisions that benefit all members of the community?’ The independent variable is which vignette treatment respondents received. Control variables include age, age squared, and sex. Participation in JASS is a binary indicator for whether a respondent participated in at least one JASS intervention. All regressions use OLS with village type fixed effects. Standard errors are clustered at the village level. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

However, we are also interested in the heterogeneous effects of being treated with

either of the two vignettes based on both allochthony status and JASS participation. Figure B1 shows the marginal effects for being treated with the community-led natural resource governance vignette; figure B2 shows the marginal effects for being treated with the state-led natural resource governance vignettes.

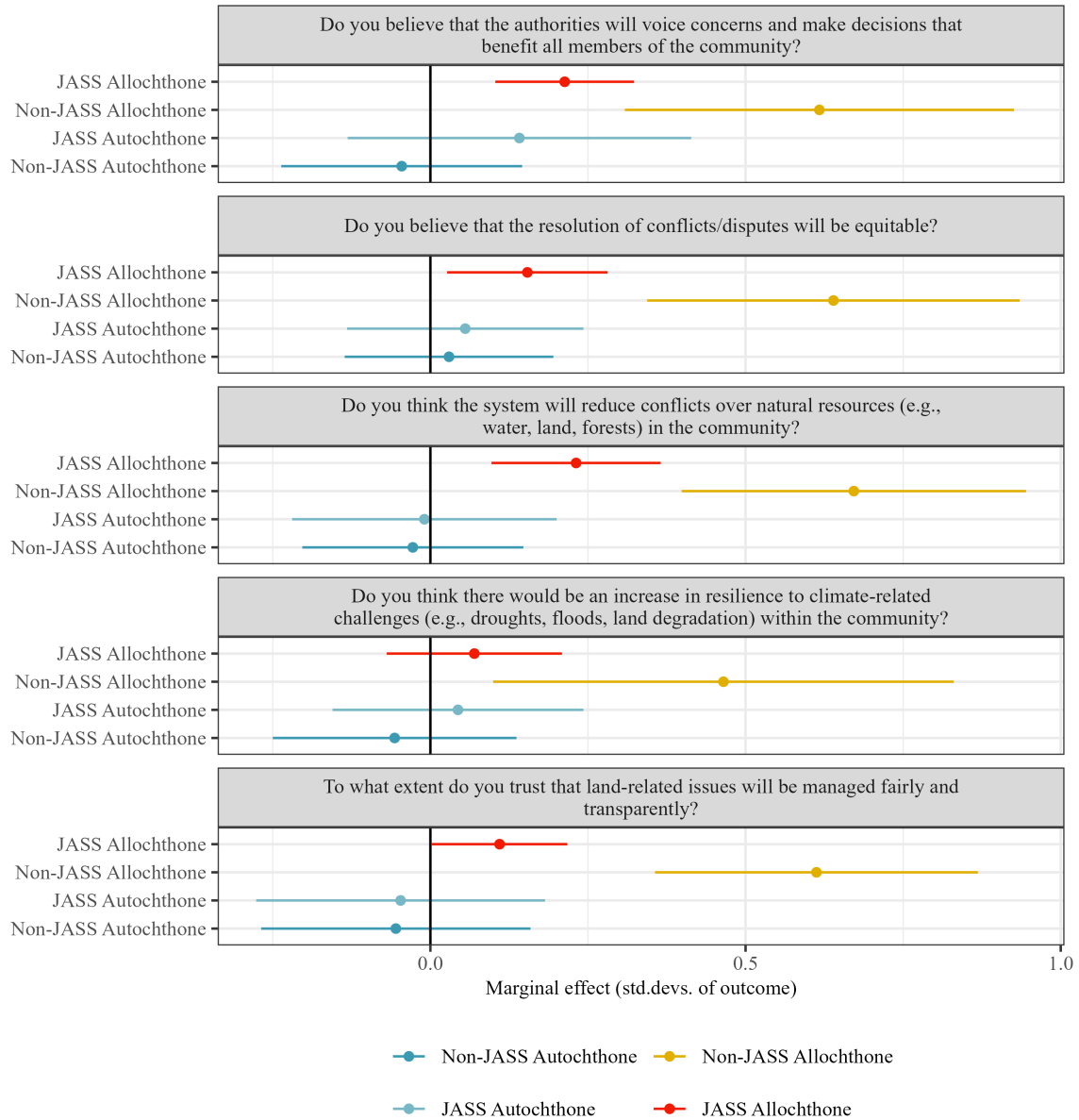
In the main text, we proxy a respondent's allochthone status using a continuous measure of the respondent's distance to the village centroid. To make the below marginal effects readable we collapse the continuous measure of allochthony into a binary measure which captures whether a respondent is above or below the median distance to a village centroid. In addition, we also collapse the multiple JASS activities in which a respondent could participate into a binary indicator for whether they participated in at least one JASS activity. Overall, 2,371 of our respondents (65.7 percent) participated in JASS. Among autochthones, 785 (52 percent) participated in at least one JASS activity while 730 did not (48 percent). Among allochthones, 1327 (87.6 percent) participated in at least one JASS activity while 188 (12.4 percent) did not.

Figure B1. Heterogenous treatment effects of being treated with the community-led vignette by allochthony status and participation in JASS



This figure shows the marginal effect of treatment with the community-led natural resource governance vignettes, expressed as standard deviations of each outcome variable by whether the respondent was an allochthone and whether they participated in at least one JASS activity. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level.

Figure B2. Heterogenous treatment effects of being treated with the state-led vignette by allochthony status and participation in JASS



This figure shows the marginal effect of treatment with the community-led natural resource governance vignettes, expressed as standard deviations of each outcome variable by whether the respondent was an allochthone and whether they participated in at least one JASS activity. All regressions use OLS with village fixed effects. Standard errors are clustered at the village level.

C Heterogeneous effects by institutional preference

In our control condition, we invite respondents to “[i]magine that your community continues to manage natural resources in the same way it has been doing for years.” However, status quo natural resource in Mali and Niger blends elements of customary or community-led governance with elements of top down, centralized governance. Consequently, by constructing our control condition in reference to a pre-existing status quo, we open space for heterogeneous treatment effects among respondents who perceive their status quo differently.

In this appendix, we explore heterogeneous treatment effects by respondents answers to ‘If you have a dispute about land, livestock, or a business transaction, who would you approach to resolve the dispute?’ We display the distribution of responses to this question in figure 1. The question appears before the experimental treatment, and allowed respondents to select multiple answers. We do not show marginal effects among respondents who said they would bring a dispute to an armed group because only 13 respondents selected that option.

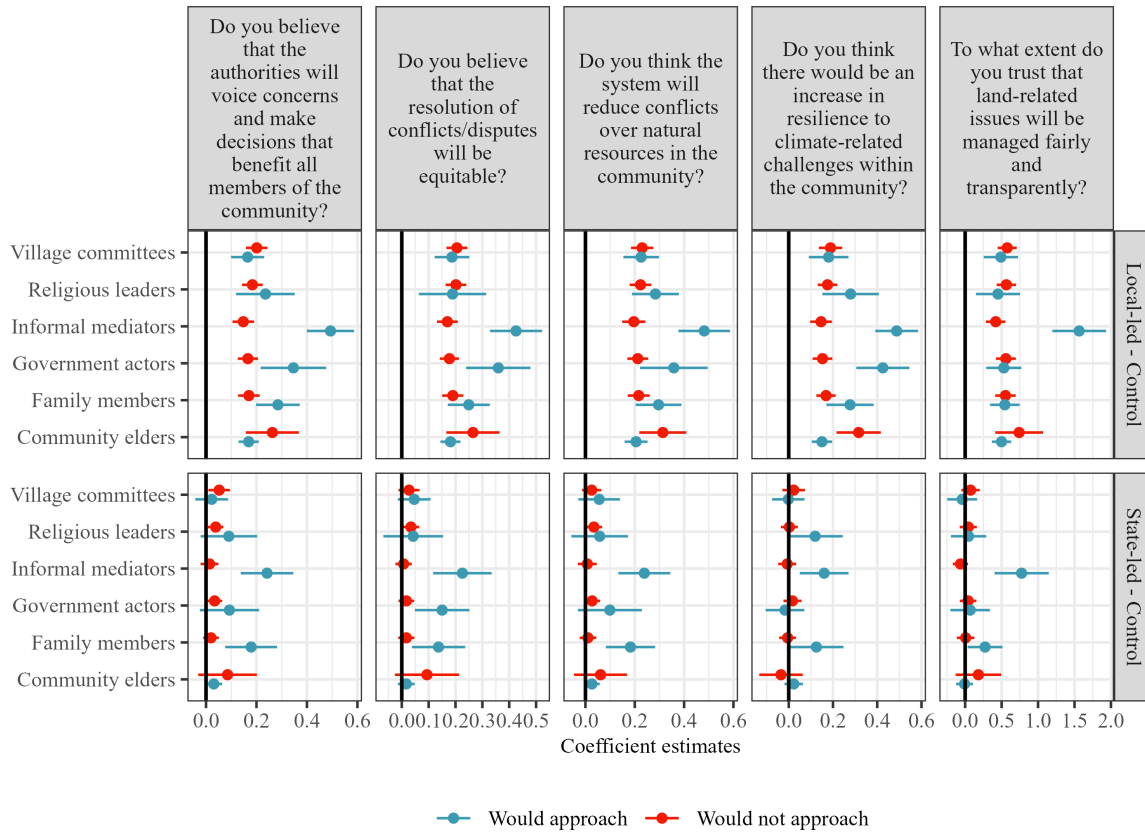
Figure C1 shows no difference among respondents who say they would approach village conflict resolution committees, religious leaders, or community elders. Among respondents who said they use informal mediators, we see much stronger effects for both the state-led and treatment led vignettes. Similarly, responses for both treatment effects are higher among respondents who say they would approach family members to resolve a conflict. One interpretation of these effects is that users of informal mediators or family mediation are uniquely unhappy with their status quo, and perceive any potential alternative to be superior. Among those who say they would approach government actors, we see higher marginal effects of receiving the community-led treatment—although no increase in ‘the extent to which [the respondent] trusts land-related issues will be managed fairly and transparently.’ They also have inconsistently higher results for the state-led vignettes.

It seems reasonable to interpret larger marginal effects for the treatment vignettes among respondents who say they would approach an institution as evidence of displeasure with this institution. If respondents like the institution, switching to a difference

body to resolve disputes wouldn't increase their confidence in dispute resolution. Following this interpretation, respondents seem less happy with body government actors and informal mediators. Among respondents who said they would approach community elders, the marginal effect of treatment is actually lower, suggesting that respondents seem to think community elders to a decent job.

However, it is also worth noting that overall, these results seem reasonably homogeneous—as least for the community led treatment. The marginal effect of receiving the vignette is consistently positive for all potential subgroups. Consequently, we feel justified in saying that different interpretations of the status quo do not drive our results.

Figure C1. Heterogenous treatment effects of being treated with the community-led vignette by allochthony status and participation in JASS



This figure shows the marginal effect of treatment with the treatment vignettes among respondents who did and did not select institutions in response to 'If you have a dispute about land, livestock, or a business transaction, who would you approach to resolve the dispute?' All regressions use OLS with village fixed effects. Standard errors are clustered at the village level.

Appendix References

Hainmueller, Jens, Jonathan Mummolo, and Yiqing Xu. 2019. “How Much Should We Trust Estimates from Multiplicative Interaction Models? Simple Tools to Improve Empirical Practice.” *Political Analysis* 27 (2): 163–192. <https://doi.org/10.1017/pan.2018.46>.